Terrain-Aware, Latency-Robust Control for Off-Road Autonomy and Teleoperation on Deformable Terrain

Kyle Sha, Zhenhao Zhou, Radu Serban, Dan Negrut
Department of Mechanical Engineering, University of Wisconsin–Madison
1513 University Ave., Madison, WI 53706, USA
Email: {kasha2, zzhou292, serban, negrut}@wisc.edu

***Abstract*—We present a modular, latency-aware framework for shared and autonomous control of ground vehicles on deformable Soil Contact Model (SCM) terrain. The central problem is integration: terrain adaptation, command screening, latency compensation, and advisory warning must operate on the same plant, message stream, and timing assumptions if their cross-layer effects are to be measured. The software is structured around six explicit extension points — command source, safety filter, collision warning, tire model, terrain estimator, and latency profile — declared through Python `Protocol` interfaces and selected by the runtime factories used by the benchmark harness. A differentiable neural tire surrogate is embedded in an *acados* nonlinear MPC (NMPC), a force/proprioceptive terrain estimator re-conditions the surrogate online from inertial and wheel-speed signals, and an asymmetric throttle disturbance observer reduces the residual soft-soil speed deficit. A two-layer obstacle-avoidance stack pairs in-horizon NMPC softplus barriers with a swappable downstream safety filter; the default filter is an intent-preserving DOB-CBF, a minimum-deviation screen suited to human teleoperation. In parallel, a terrain- and latency-aware forward collision warning re-uses the same tire surrogate to derive a soil-conditioned braking budget and advise the operator without overriding commands. Chrono::Vehicle inside Chrono::HIL provides the plant, and a synthetic 5G latency trace — from a sequence model fit to a public 5G uplink dataset and mapped through a queue-load delay model — drives the configurable command and camera delays. Repeatable simulation experiments show that the terrain-aware neural stack reduces median RMS crosstrack error by 37–$38\,\%$ relative to SCM-calibrated Pacejka and TMeasy baselines in the live-estimator regime — a gain concentrated in off-nominal (dirt, higher-speed) cells where the fixed analytical models lose grip, with medians otherwise comparable — the disturbance observer increases achieved speed with a small tracking penalty, the DOB-CBF filter reduces collisions under the synthetic 5G profile, a deterministic counterfactual replay attributes 31 of 39 matched averted collisions to the filter under scripted worst-case drive-into-hazard commands, and the warning reproduces Chrono brake-stop distances to within $0.17\,\mathrm{m}$ mean absolute error.**

***Index Terms*—Off-road teleoperation, soil contact model, neural surrogate, model predictive control, control barrier function, safety filter, 5G latency, Chrono.**

I. Introduction

Off-road teleoperation must contend with two compounding sources of uncertainty: time-varying soil mechanics that rigid-terrain tire models cannot capture without per-deployment recalibration [3], [1], and time-varying network latency that destabilises naive shared-control arbitration [2], [8]. Existing approaches address these in isolation: terrain-adaptive trajectory planning typically assumes nominal communication, while delay-compensated safety filters typically assume rigid-terrain dynamics. This paper integrates the two into a single closed-loop framework and reports paired simulation studies that isolate the main cross-layer design choices.

The framework spans four coupled layers around a single deformable-terrain plant: (i) a terrain-aware NMPC built on a differentiable neural tire surrogate; (ii) an online terrain estimator that re-conditions that surrogate from onboard sensors; (iii) a two-layer obstacle-avoidance stack (in-horizon barriers + swappable downstream safety filter) for both autonomous and teleoperated commands; and (iv) a terrain- and latency-aware forward collision warning that re-uses the same tire surrogate to provide an advisory channel when an override-style filter is undesirable. All four layers share the same Chrono::Vehicle plant inside Chrono::HIL, the same ZeroMQ messaging substrate, and the same benchmarking harness, so the main ablations are comparable within one controlled experimental matrix.

Concretely, this integration paper contributes:

1) A modular framework (Section II) in which six runtime roles are declared as Python `Protocols` with corresponding registry interfaces. The registry-instantiated safety-filter and warning layers are covered by a runtime conformance test, while command-source, tire-model, and terrain-estimator variants retain the established CLI/factory selection paths used by the benchmark suite.
2) A differentiable neural SCM tire surrogate (Section III) that augments the static MLP of Dallas et al. [3] with rate-aware and axle-rate-aware variants, exported as a CasADi expression through which the acados NMPC auto-differentiates at every horizon stage.
3) A sliding-window online terrain estimator (Section IV) that recovers Bekker n from inertial and wheel-speed signals alone and re-conditions the surrogate over the controller-to-sim socket. The deployed estimator fuses a learned lateral-force channel with a proprioceptive window-MLP inside a state-augmented UKF [3]; on a 100-soil sweep along the clay–dirt–sand Bekker–Mohr

manifold with the other five Bekker–Mohr parameters held near their presets (Chrono SCM, live closed-loop NMPC), it tracks n across the deployable range with a $4.5\,\%$ median $|\Delta n|/n_{\mathrm{true}}$ ($94\,\%$ of soils within $\pm 20\,\%$) — the best of four backends, and well ahead of the force-only UKFs, which saturate on firm soils once closed-loop slip is small.

4) A terrain-aware speed planner (Section V) that turns the surrogate's grip limits at the live $\hat{n}$ into a friction-circle (g–g) speed profile, cutting closed-loop tracking error by $51\,\%$ over a curvature-only reference by slowing only where the soil cannot support the corner.
5) An asymmetric throttle disturbance observer (Section VI) that closes the residual soft-soil speed gap left by the open-loop longitudinal integrator.
6) A two-layer obstacle-avoidance stack (Section VII) pairing in-horizon NMPC softplus barriers with a swappable downstream safety filter; the default instance is an intent-preserving DOB-CBF-QP inspired by [8], with physical steering/throttle rate limits imposed directly in the QP. Its effect is quantified by a deterministic *counterfactual replay* — replaying a fixed command trace (filter off vs. on) on the identical scenario — giving a paired, reproducible collision-prevention measurement over a scripted population of worst-case drive-into-hazard commands (a proxy for an inattentive or latency-overwhelmed operator).
7) A synthetic 5G latency model (Section VIII): a sequence model fit to a public 5G uplink dataset, mapped through a queue-load delay on both the operator-command and camera paths, exercising closed-loop robustness (the queue-load mapping and outage spikes are modeling choices, not measured link behavior).
8) A terrain- and latency-aware forward collision warning (Section IX) that re-uses the rig tire surrogate for a soil-conditioned braking budget, advises without override, and matches actual Chrono brake stops to $0.17\,\mathrm{m}$ mean absolute error.
9) An open-source benchmarking suite (Section X) that reproduces every non-HIL figure and CSV from a single command, and an integrated end-to-end demonstration (Section XI) that runs all layers together on one mission — a proof-of-concept that the components compose in an end-to-end run (a single mission, not a statistical full-stack claim; see Section XI).

Several of these components individually extend prior work incrementally. The contribution of this paper is their *integration* and head-to-head evaluation on one deformable-terrain plant, one messaging substrate, and one benchmarking harness — which is what exposes the cross-layer interactions that isolated studies cannot, such as the safety filter re-using the controller's tire surrogate for its traction-budget rollouts and the collision warning re-using the same soil-conditioned braking budget. Terrain adaptation and delay-compensated safety have previously been studied separately; here they are measured together on the same matrix.

TABLE I: The six extension points of the framework. Each role is defined by one Protocol and has a corresponding registry interface; the current runtime conformance test exercises the registry-wired safety-filter and collision-warning roles.

Role	Protocol method	Current variants
Command source	`next_command`	NMPC, G29, WASD
Safety filter	`filter`	DOB-CBF; no-filter bypass
Collision warning	`evaluate`	TTC
Tire-force model	`predict`	rate-MLP, axle-rate-MLP, Pacejka, TMeasy
Terrain estimator	`observe/estimate`	force+proprioceptive UKF, window-MLP, Bekker/NN-UKF
Latency profile	`delay`	constant, replay, learned 5G

II. System Architecture

The runtime is organised around six explicit extension points (Fig. 1, Table I). Each extension point is declared as a Python protocol with a corresponding registry interface. In the current implementation the safety filter, collision warning, and latency profile are registry-instantiated; command-source, tire-model, and terrain-estimator variants use the established CLI/factory paths while satisfying the same protocol boundaries. Chrono::Vehicle simulates an HMMWV on a Soil Contact Model (SCM) [4] patch within Chrono::HIL [11]. The simulation and controller nodes run as separate operating-system processes and exchange messages over ZeroMQ pub/sub sockets: a vehicle-state message at the physics state rate ($100\,\mathrm{Hz}$), a control command at the controller rate ($10\,\mathrm{Hz}$), and a lifecycle status message. The control command carries the most recent live terrain estimate as optional fields so the sim-side safety filter re-conditions itself without a separate high-priority channel; the deployed estimator is n-only, and the remaining Bekker–Mohr fields are mapped from $\hat{n}$ along the terrain manifold before attachment. Piggy-backing is required because the controller-to-simulation socket is latest-only (conflating), which would drop a separate terrain-update message arriving between estimator ticks.

Both processes pace to wall clock, and the benchmark harness parallelises across runs, so a single workstation completes the static-terrain tire benchmark of Section III-B in well under an hour.

The split between process-boundary contracts (ZMQ messages) and in-process Protocol contracts is deliberate. Any controller that subscribes to the vehicle-state message and publishes the control command can replace the acados NMPC without modifying the simulator; this is how the tire-model variants of §III are evaluated. Human teleoperation is reduced to the same normalised steering, throttle, and brake command channels before actuation, so the safety layer screens either autonomous or human commands behind one contract. Safety filters and the forward collision warning live behind their own Protocols, are resolved by name through their registries, and never share state — the warning attaches to the same vehicle-state stream but never writes to the command path, so it composes with any selected filter (including the no-filter case)

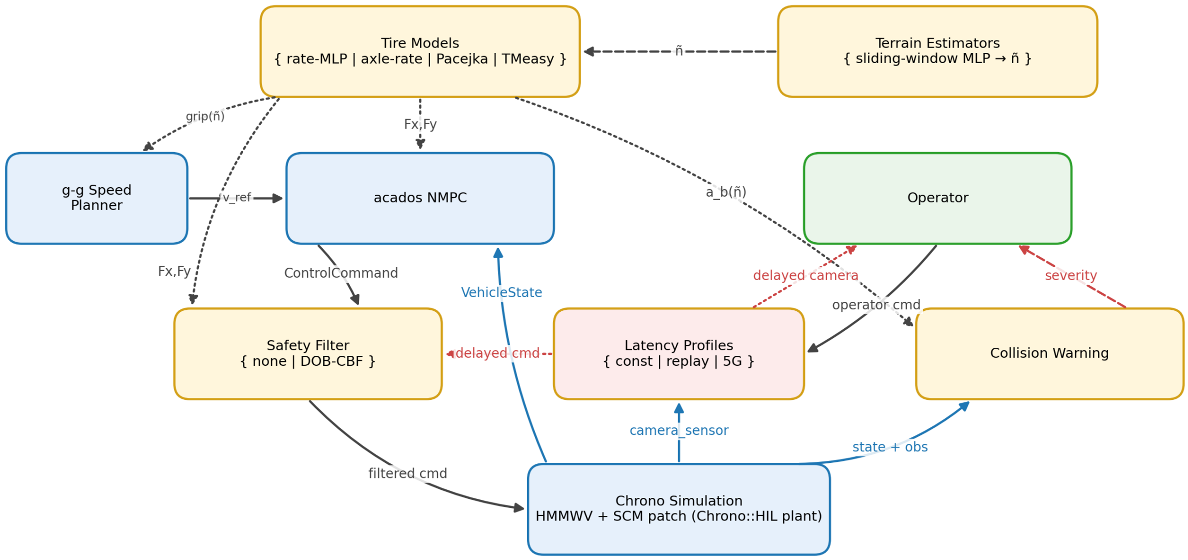

Fig. 1: Shared-control framework. Yellow nodes are in-process extension points selected by the runtime factories; blue/green nodes are process-boundary or operator endpoints. Solid arrows are runtime data flow over ZeroMQ; dashed arrows are the latency-affected operator paths; dotted arrows are tire-surrogate queries (build-time CasADi export plus per-tick safety-filter rollouts).

without re-plumbing. This is what makes the suite a framework rather than a single monolithic controller demonstration.

III. Differentiable SCM Tire Surrogate

A multilayer perceptron (MLP) maps the tire operating point (κ, α, F_z, u), steering rate, and soil parameters $\boldsymbol{\theta}_{\text{soil}} = (K_\phi, K_c, n, c, \phi, k)$ (Bekker–Wong-style rig inputs [6]) to longitudinal and lateral tire forces (F_x, F_y). Where Dallas et al. [3] use a static MLP only, we train three variants: the static baseline, a *rate-augmented* MLP that adds the steering rate $\dot{\delta}$ as an explicit input, and an *axle-rate* MLP with per-axle rate inputs $(\dot{\delta}_f, \dot{\delta}_r)$. The rate input lets the surrogate represent the tire force while the operating point is *changing* — the transient during fast steering — rather than only the quasi-steady force a static map returns; in practice (Sec. III-D) this mainly buys robustness against a transient failure mode rather than a bulk accuracy gain. All three export as CasADi expressions through which the acados solver auto-differentiates at every horizon stage.

Two surrogate *generations* are kept distinct, and the distinction recurs throughout the paper. The *tire-rig surrogate* is trained from controlled single-tire SCM sweeps — one tire on a virtual test rig, stepped over a grid of $(\kappa, \alpha, F_z, \boldsymbol{\theta}_{\text{soil}})$ with the exact SCM force logged at each operating point (the reproducible protocol of [3]). The *whole-vehicle surrogate* is trained from closed-loop Chrono::Vehicle HMMWV runs that log each tire's live operating point alongside the SCM ground-truth force, so it captures the coupled regimes a driven vehicle actually reaches — load transfer, yaw, and the body-frame force conventions the isolated rig never produces. The rig generation tests whether the single-tire learning problem is solved in isolation; the whole-vehicle generation (deployed for the main NMPC and safety-filter results) tests whether the controller sees accurate forces in the regimes it actually drives through.

A. Training data: from rig sweeps to closed-loop rollouts

The training-data pipeline followed two generations. The first generation, mirroring the rig protocol of [3], drove a single SCM tire over a Cartesian grid of $(\kappa, \alpha, F_z, \boldsymbol{\theta}_{\text{soil}})$ and recorded the SCM ground-truth force at every queried operating point. Rig training is appealing because every label is exact, no closed-loop feedback confounds the supervision, and the data collection parallelises trivially. It produced surrogates with rig-grid coefficient of determination $\mathrm{R}^2 \gtrsim 0.95$ on a held-out rig split, and it is therefore retained as the controlled baseline. The second generation targets the controller's deployed operating distribution: rather than sampling an open-loop tire grid, it collects training tuples from the closed-loop stack itself so the labels include the high-yaw-rate, lateral-acceleration, and body-frame force conventions that arise during aggressive references on deformable soil.

For the closed-loop generation, a parallel pool of randomised (terrain, path, speed, bumpiness, seed) scenarios runs the NMPC against Chrono::Vehicle and logs the live operating point alongside the SCM ground-truth force. The deployed whole-vehicle surrogate `vehicle_rate_64_32_lhs` is trained on **800 such scenarios**, LHS-sampled jointly over the soil parameters in `TRAINING_RANGES_V6` (Bekker–Mohr box around clay/dirt/sand) and the maneuver axes — 264 sand, 270 dirt, 266 clay; 214 sinusoidal, 201 right-left, 200 lane-change, 185 double-lane-change; commanded speeds randomised in $[3, 7]\,\mathrm{m/s}$; bumpiness and rock-count randomised; each scenario simulated for $\sim$10 s at the controller rate. The aggregated training table contains approximately $1.85\,\mathrm{M}$ rows of $(\kappa, \alpha, F_z, u, \dot{\delta}) \rightarrow (F_x, F_y)$ tuples (front + rear axle per tick) — fewer samples than the rig's $\sim$100k single-tire grid but drawn from the operating-point distribution the deployed MPC actually visits. The resulting distribution is matched to the regime the planner actually visits, and the body-frame convention is enforced at the collection step rather than at the import step. The three checkpoints used throughout this paper's pilot sweeps are all closed-loop trained; two rig-trained checkpoints from the first generation are retained as a baseline so the rig-versus-closed-loop comparison remains reproducible (Sec. III-B selects them as the `rig_static` and `rig_rate` model identifiers).

The NMPC queries the surrogate at each stage. The longitudinal traction budget F_x^{trac} is evaluated at reference slip κ_{ref}, $\alpha = 0$, with axle loads F_z, speed u, and $\boldsymbol{\theta}_{\mathrm{soil}}$ taken from terrain presets or the online estimator. Cornering forces F_y use the live operating point (κ, α, F_z, u) at each stage.

The two *learned terrain estimators* are trained on deliberately different data, and the distinction matters for reading their results. The sliding-window MLP (`terrain_window_mlp`, Sec. IV) — the proprioceptive channel of the deployed Fused-UKF (Sec. IV-B) — regresses the Bekker exponent n from a 4 s window of proprioceptive statistics; it is trained on 180 LHS soil cells $\times$ 20 scenarios, split between closed-loop NMPC tracking (sinusoidal / lane-change / right-left at $\{4, 6, 8\}\,\mathrm{m/s}$) and scripted open-loop cruise-bias runs, at bumpiness $\in \{0, 4\}$ only — the envelope that bounds its in-distribution claims (Table II). The NN-UKF tire surrogate (`vehicle_fy`, Sec. IV-B) is instead trained on 300 *scripted* sinusoidal-steer runs with no controller in the loop, half at constant open-loop throttle and half under a PI cruise loop, over a widened soil box so the canonical presets are interior points. The throttle convention thus differs across models — the deployed surrogates see NMPC cruise throttle, the rig surrogate sees none (fixed rig), and the UKF surrogate sees a mix of constant-open-loop and PI-cruise throttle — so each model's accuracy is reported only within its own excitation envelope.

B. Static-terrain benchmark

The standard MPC is swept over its three deployable tire-force models (the analytical Pacejka and TMeasy, and the deployed vehicle NN rate-MLP surrogate) across all three terrains (clay, dirt, sand), three reference paths (sinusoidal, lane-change, right-left), three reference speeds $v_{\mathrm{ref}} \in \{5, 7, 9\}\,\mathrm{m/s}$, three bumpiness levels, and five random seeds, for a total of 1215 closed-loop trials. Sensor noise is enabled in every trial. To keep the comparison fair, the analytical baselines are *not* run at rigid-road defaults: their peak lateral friction is calibrated to a single global $\mu = 0.42$ by least-squares fit to the pooled SCM tire-rig forces across the soil box (the SCM peak $|F_y|/F_z$ ranges from ≈ 0.18 clay-like to ≈ 0.47 sand-like, far below a rigid road), with the magic-

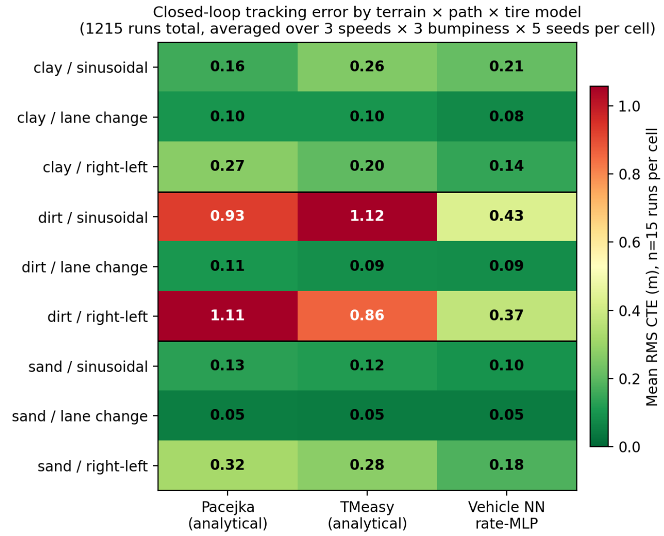

Fig. 2: Closed-loop RMS CTE per (terrain, path) cell across the broader 1215-run sweep (3 speeds $\times$ 3 bumpiness $\times$ 5 seeds per cell). The analytical baselines (Pacejka, TMeasy) track well on clay and sand but diverge on dirt/sinusoidal ($\sim$1.0 m) and dirt/right-left (0.9–1.2 m), where the curvature-limited reference combines with off-nominal sinkage to produce a regime the analytical force laws do not cover even after SCM calibration. The Vehicle NN rate-MLP suppresses those tails to 0.4–0.6 m while matching the calibrated baselines on the benign cells. This per-scenario view is what motivates the live-estimator benchmark in Sec. III-C.

formula and TMeasy shape factors kept at their standard values; one global set is used for every terrain because the controller has no per-terrain knowledge. This removes the rigid-road over-grip that otherwise inflates the baseline error, so the reported gap reflects what a fixed analytical parameterisation structurally cannot capture, not a tuning deficit. Table III reports the mean and standard deviation of root-mean-square crosstrack error (RMS CTE), speed retention ratio, and solver wall time across the matrix. The neural row is the deployed whole-vehicle surrogate; the rig-trained checkpoints are retained in the benchmark suite as explicit `rig_static` and `rig_rate` baselines for reproducing the rig-to-vehicle comparison. Fig. 2 breaks the same matrix out per (terrain, path), which makes the analytical-baseline failure mode at dirt/sinusoidal and dirt/right-left visible at a glance.

C. Live-estimator-conditioned benchmark

The stronger claim, that the neural surrogate combined with the online terrain estimator improves tracking over Pacejka and TMeasy, requires the estimator to update the soil parameters that the surrogate consumes. The same model set is therefore re-evaluated with the online estimator enabled, using a 20-second simulation window with key performance indicators (KPIs) computed after an 8-second post-convergence cutoff. Table IV reports the result; Fig. 3 shows the per-scenario heatmap. Because this regime conditions on the live estimator

TABLE II: Training-data provenance for the learned models. "Loop" distinguishes scripted open-loop excitation from closed-loop NMPC tracking; bumpiness is the terrain-roughness level present during training (and hence the envelope within which each model is evaluated).

Model (role)	Loop / excitation	Throttle	Bumpiness	Soil / n coverage
`rig_rate` (rig tire baseline)	open-loop single-tire rig, scripted κ/α	— (fixed rig)	—	LHS around clay/dirt/sand
`vehicle_rate` (deployed NMPC tire)	closed-loop NMPC, 4 paths	NMPC cruise, $[3, 7]$ m/s	$\{0, 2, 4\}$	full Bekker–Mohr LHS box
`terrain_window_mlp` (window-MLP channel of deployed Fused-UKF)	closed-loop NMPC + scripted open-loop	NMPC + cruise-bias	$\{0, 4\}$	$n \in [0.40, 1.30]$, 180 LHS cells
`vehicle_fy` (NN-UKF tire, Sec. IV-B)	scripted sinusoidal steer, no controller	half const. OL / half PI-cruise	0	widened $n \in [0.35, 1.35]$

TABLE III: Tire-model benchmark over the full 1215-run matrix (405 runs per model), against *SCM-calibrated* analytical baselines (single global $\mu = 0.42$, Sec. III-B). The deployed vehicle NN rate-MLP attains the lowest mean RMS CTE, 46–48 % below the analytical baselines; the per-model *medians* are close (0.12–0.15 m), so the NN's gain is in the tail — the Pacejka and TMeasy rows carry large right-skewed standard deviations (0.44–0.45 vs. 0.17) from off-nominal dirt scenarios (Fig. 2) that the NN suppresses. All models retain similar speed because this controlled comparison fixes the speed reference to the geometric curvature profile across every tire model (the deployed terrain-aware speed planner of Sec. V is held out so the comparison isolates the tracking tire model), so the reference, not the tire force model, binds speed here. The NN's higher solve time reflects the embedded surrogate evaluation; all three rows remain below the 100 ms controller period.

Tire model	RMS CTE (m)	Speed ratio	Solve (ms)
Pacejka	0.35 ± 0.45	0.66	9.7
TMeasy	0.34 ± 0.44	0.65	10.1
Vehicle NN rate-MLP	$\mathbf{0.18 \pm 0.17}$	0.67	28.4

(not the static terrain prior of Table III) and reports *medians* over the larger 1620-run matrix, its analytical-baseline numbers are not directly comparable to Table III's mean±std over the 1215-run static matrix; the two report the static-prior and live-estimator regimes respectively.

Two points are needed to read Table IV honestly. First, the `static` neural row is conditioned on the matched terrain preset and therefore behaves as a near-oracle reference, whereas the `+estimator` row must recover the soil parameters online from inertial and wheel-speed signals alone. The estimator variant nearly matches that near-oracle row (median 0.177 vs. 0.174 m, a 3 mm gap whose 95 % bootstrap confidence intervals, $[0.150, 0.191]$ and $[0.142, 0.191]$ m, overlap — statistically indistinguishable) while requiring no prior terrain knowledge, and both NN rows reduce median RMS CTE by 37–38 % relative to the SCM-calibrated Pacejka and TMeasy baselines. The deployable claim is therefore that the estimator recovers most of the oracle-prior benefit without that prior,

TABLE IV: Live-estimator-conditioned tire-model benchmark, full 1620-run scheduled matrix (405 per variant), KPI window from $t = 8$ s. Values are *median* RMS CTE: the per-variant distributions are strongly right-skewed by a few high-error dirt scenarios (Fig. 3), so the median is the representative statistic. The `static` neural row is conditioned on the matched terrain preset (near-oracle); the `+estimator` row runs the deployed force+proprioceptive Fused-UKF (Sec. IV-B) online and nearly matches the oracle (median 0.177 vs. 0.174 m, a 3 mm gap with overlapping 95 % bootstrap CIs). Both NN rows reduce median RMS CTE by 37–38 % relative to the *SCM-calibrated* analytical baselines (single global $\mu = 0.42$).

Variant	CTE (m)	vs. Pacejka	vs. TMeasy
Pacejka (static)	0.28	—	—
TMeasy (static)	0.28	—	—
NN rate-MLP (static)	0.17	−38 %	−38 %
NN rate-MLP + estimator	**0.18**	**−37 %**	**−37 %**

not that it improves on a correct prior. Second, the per-row distributions are strongly right-skewed by a few high-error scenarios (mostly Pacejka and TMeasy on dirt at $v_{\text{ref}} = 7$ m/s, where the analytical baselines collide with the curvature limit); Fig. 3 shows the full distribution per variant so the median and tail behaviour are both visible.

A separate ablation isolates the deployable value of the estimator explicitly. Holding the controller's static terrain prior at dirt regardless of the plant terrain gives the wrong-prior baseline, comparable directly against the matched-prior oracle and the estimator-on variant on the same matrix (clay+sand $\times$ sinusoidal $\times$ $v \in \{5, 7\}$ m/s $\times$ flat terrain $\times$ 5 seeds, 20 runs per variant). Table V reports the result. On clay the wrong dirt prior gives RMS CTE 1.56 m, the estimator-on variant (the deployed Fused-UKF, Sec. IV-B) adapts from the same dirt initialisation and recovers to 0.68 m, close to the matched-prior oracle (0.64 m) – the estimator closes $\approx$ 95 % of the prior-mismatch gap on the harder terrain.

D. Controlled rig-versus-whole-vehicle comparison

The rig-vs-whole-vehicle question motivating Sec. III-A is settled by a controlled retrain that removes two confounds present in the earlier comparison: rig checkpoints used smaller MLPs (16-4, 16-8) than vehicle checkpoints (32-16), and

TABLE V: Wrong-prior ablation: RMS CTE (m) when the controller's static terrain prior is fixed to dirt regardless of plant terrain, compared against the matched-prior near-oracle and the estimator-on variant. 60 runs total (20 per variant). The matrix excludes dirt (where the dirt-prior is by construction matched) and the other paths/bumpiness levels of Table IV; absolute numbers are therefore not comparable across the two tables. The estimator nearly recovers the matched-prior performance from a wrong initialisation on clay; matched and wrong priors both yield low error on sand because dirt is closer to sand than to clay on the soil-parameter manifold.

Variant	Clay	Sand	Mean
Matched static prior (oracle)	0.641	0.056	0.348
Wrong (dirt) static prior	1.562	0.063	0.813
Estimator-on (dirt → adapt)	**0.684**	**0.062**	**0.373**

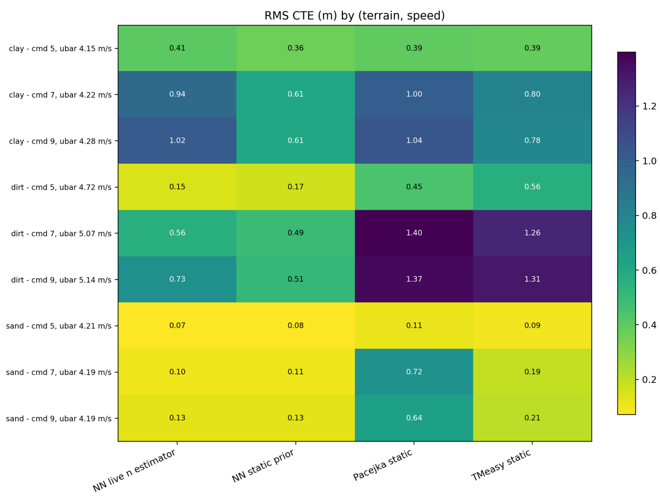

Fig. 3: Same benchmark, RMS CTE distribution per variant. Box plots show interquartile range and median across the valid KPI rows from the 405 scheduled runs per variant; jittered scatter points are coloured by terrain (clay, dirt, sand). The two analytical baselines (Pacejka, TMeasy) carry long right tails driven by a few dirt cases at higher v_{ref}; both neural variants substantially compress the distribution tail. The estimator-on variant nearly matches the matched-prior static row, the deployable claim of Sec. III-C.

the vehicle training set jittered scenarios around the three canonical deployment terrains rather than sampling broadly. We retrain:

- three rig checkpoints (32-16 static, 32-16 rate, 64-32 rate) on the existing 100k single-tire SCM CSVs;
- three whole-vehicle checkpoints (matched architectures) on a fresh 800-scenario closed-loop dataset that LHS-samples $\boldsymbol{\theta}_{\mathrm{soil}}$ across the same Bekker–Mohr box the rig uses (`TRAINING_RANGES_V6`; no canonical-preset jitter).

Offline force prediction on a held-out LHS test slice exposes the generalisation gap that the canonical-jitter eval set previously hid: the legacy whole-vehicle checkpoints' $R^2(F_y)$ drops from $+0.58$ on canonical-jitter data to -0.30 on LHS-terrain data, while the LHS-trained replacements hold $R^2 \approx 0.71$ on both eval sets. The retrained vehicle MLPs were not better at predicting in-distribution data; the legacy MLPs had been over-fitting the deployment-terrain neighbourhood.

TABLE VI: Controlled rig-vs-whole-vehicle sweep (432 closed-loop runs, planner-only, sensor noise on). Matched architecture in each pair; rig retrains use the kept 100k LHS rig CSV, vehicle retrains use a fresh 800-scenario LHS-terrain closed-loop dataset. The earlier small-capacity rig and vehicle checkpoints are preserved in the benchmark suite for reproducibility.

Surrogate (32-16 / 64-32 paired)	RMS CTE (m)	Speed ratio	Solve (ms)
Rig static (32-16)	**0.095**	0.73	10.5
Vehicle static (32-16)	0.118	0.73	10.7
Rig rate (32-16)	0.099	0.73	12.5
Vehicle rate (32-16)	0.096	0.72	12.5
Rig rate (64-32)	0.103	0.73	16.7
Vehicle rate (64-32)	0.106	0.73	15.6

The controlled closed-loop tire-model sweep (432 runs, 6 surrogates × 3 terrains × 3 paths × 2 speeds × 2 bumpiness × 2 seeds; planner-only, sensor noise on) is summarised in Table VI. The pre-retrain rig-trained advantage narrows substantially:

- Static MLPs: the rig surrogate remains decisively better ($0.095\,\mathrm{m}$ vs $0.118\,\mathrm{m}$, $20\,\%$).
- Rate MLPs (32-16): rig and vehicle match (0.099 vs $0.096\,\mathrm{m}$).
- Rate MLPs (64-32): tied within noise (0.103 vs $0.106\,\mathrm{m}$).

Per-tick force-prediction smoothness explains the shrinking gap: aggregated over the 432 runs, the LHS-trained vehicle surrogates have $\mathrm{std}(\dot{F}_y^{\mathrm{pred}})$ around 16-20 kN/s, much closer to the rig surrogates' 14 kN/s (the 32-16 rig_rate variant) than the legacy vehicle MLPs' 40 kN/s. Wider training-terrain coverage acts as an effective low-pass on the closed-loop SCM signal in the same way LHS-independent sampling did for the rig.

a) Force-RMSE vs control-RMS-CTE picture is not the same metric.: The CTE numbers in Table VI are within 1 cm of each other across the four mid-size variants. To unpack what each surrogate is doing during those runs, we replay all four surrogates on twelve logged trajectories (one of which is shown in Fig. 5), computing per-tick front-axle $|\Delta F_y|$ against the Chrono ground truth without re-running the simulator. Mean RMSE across the twelve scenarios:

Surrogate	Mean RMSE F_y (N)
Rig static (32-16)	632
Vehicle static (32-16)	645
Rig rate (64-32)	671
Vehicle rate (64-32, deployed)	706
Ensemble (0.4 rig + 0.6 vehicle, rate)	637

The deployed vehicle rate surrogate is the *worst* of the four on this force-RMSE metric, yet it is also the most reliable on closed-loop tracking. A six-scenario CTE comparison driving the standard MPC with each surrogate shows the static and

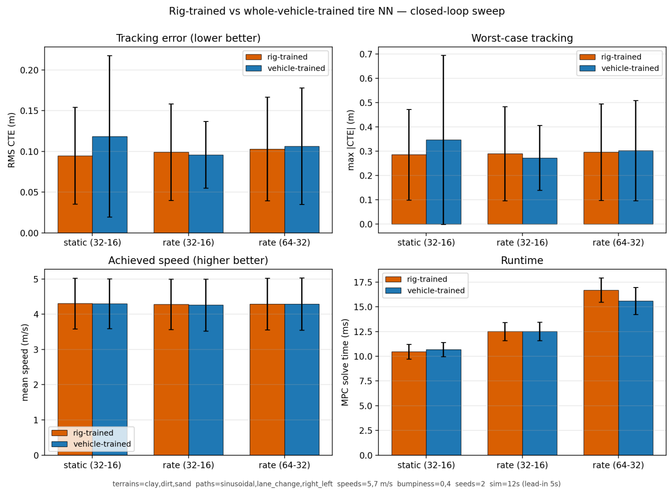

Fig. 4: Aggregate metrics over the 432-run controlled sweep, six surrogates paired by architecture and signature. RMS CTE, worst-case $|\mathrm{CTE}|$, achieved speed, and solver wall time on the same matrix. Bars are mean $\pm$ s.d. across 72 cells per surrogate. The pre-retrain rig-trained advantage narrows to a static-only effect once architecture is matched and the whole-vehicle training data is LHS-sampled over the same Bekker–Mohr box the rig uses.

rate vehicle surrogates within 5 cm of each other on five of six scenarios — and a single catastrophic failure on clay/right-left at $v_{\mathrm{ref}} = 7\,\mathrm{m/s}$ where the static surrogate degrades to $1.07\,\mathrm{m}$ RMS CTE while the rate surrogate stays at $0.16\,\mathrm{m}$. The mean RMS CTE across the six scenarios is therefore $0.18\,\mathrm{m}$ for the rate variant and $0.33\,\mathrm{m}$ for the static variant; excluding the catastrophic case they match at $0.185\,\mathrm{m}$. The rate-augmented variants trade their force-RMSE deficit for a lower-bandwidth, lower-amplitude F_y trace whose under-prediction at transient peaks (the controller that collected the LHS data planned conservatively, so the training distribution underweights extreme-slip excursions) is compensated by a smoother MPC control output and an absence of the catastrophic failure mode the static checkpoint exhibits. A simple $w = 0.4$ rig-plus-vehicle ensemble cancels the opposing biases at zero deployment cost (RMSE drops to 637 N, a 5 % improvement over vehicle alone), and is the recommended default for downstream force-prediction consumers such as the safety filter and the collision-warning braking budget.

The takeaway for deployment: the whole-vehicle pipeline is the right default *when the training data is LHS-sampled across the same operating-point box the rig spans*. Replacing the legacy canonical-preset jitter with the LHS-terrain dataset closed the generalisation gap that the rig pipeline previously exposed; the default DOB-CBF safety filter, which reads the surrogate as a static traction-ceiling query, pairs cleanly with either surrogate. The full retrain log – the offline force-prediction tables on both evaluation sets and the complete smoothness aggregation – is omitted here for space.

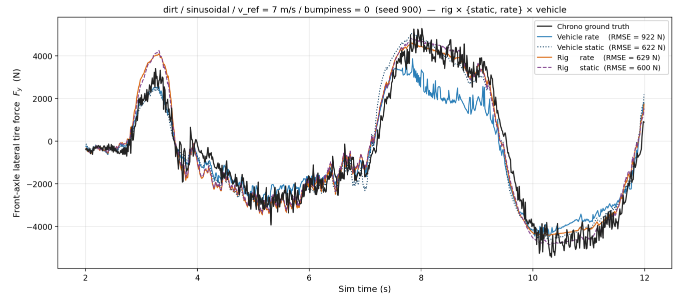

Fig. 5: One representative scenario from the four-way comparison (dirt, sinusoidal, $v_{\mathrm{ref}} = 7\,\mathrm{m/s}$, $b = 0$), front-axle F_y. Black is Chrono ground truth; the four coloured traces are the deployed vehicle rate surrogate (light blue), the vehicle static surrogate (dark blue dotted), rig rate (orange), and rig static (purple dashed). Vehicle rate consistently under-predicts the peaks; the rig variants over-predict the same peaks; both static checkpoints track the envelope most tightly. The deployed default still attains the lowest RMS CTE but at the cost of the largest force-RMSE.

E. Open-loop prediction validation

The closed-loop CTE results establish that the surrogate is accurate enough to control with, but not whether the NMPC's internal predictions match the plant — the property an MPC relies on. We validate this directly: the controller logs its full predicted horizon trajectory (4 s at 0.1 s stages) at every solve, and we compare the predicted state at each horizon stage against the actual Chrono trajectory that many steps later, aggregated over 18 runs (clay/dirt/sand $\times$ sinusoidal $\times\, v \in \{5, 7\}$ m/s $\times$ 3 seeds, deployed vehicle-rate surrogate). Fig. 6 overlays the controller's predicted horizons directly on the actual plant trace for a representative firm-sand run; the per-terrain drift magnitudes quoted below are means over the full matrix.

The NMPC predicts the *path geometry* accurately — heading drift stays below 0.15 rad over the full 4 s horizon on every terrain — but *over-predicts longitudinal speed* by $\approx$0.5 m/s almost immediately, growing to $\approx$1.0 m/s by 4 s on firm sand. The position drift (up to 2.7 m at 4 s) is therefore dominated by accumulated speed error, not lateral error. The prediction weakness is concentrated in the longitudinal channel, and a signed analysis shows the bias is *terrain-dependent*: firm soil over-predicts speed, soft dirt under-predicts, and clay is accurate. It reflects the kinematic longitudinal model ($\dot{u} = a_x$) omitting a terrain-varying motion-resistance term — not a tire-surrogate feature deficit, since the surrogate F_x is not used in the longitudinal channel (Sec. III-F). The throttle disturbance observer (Sec. VI) closes this gap online and the 10 Hz re-planning masks it in closed loop; the lateral (F_y) channel is already accurate.

F. Feature headroom and the structural force-prediction floor

Section III-E isolates F_x as the dominant prediction error. We test directly whether that gap, and the companion rear-axle F_y weakness, is closable with additional *causal, MPC-*

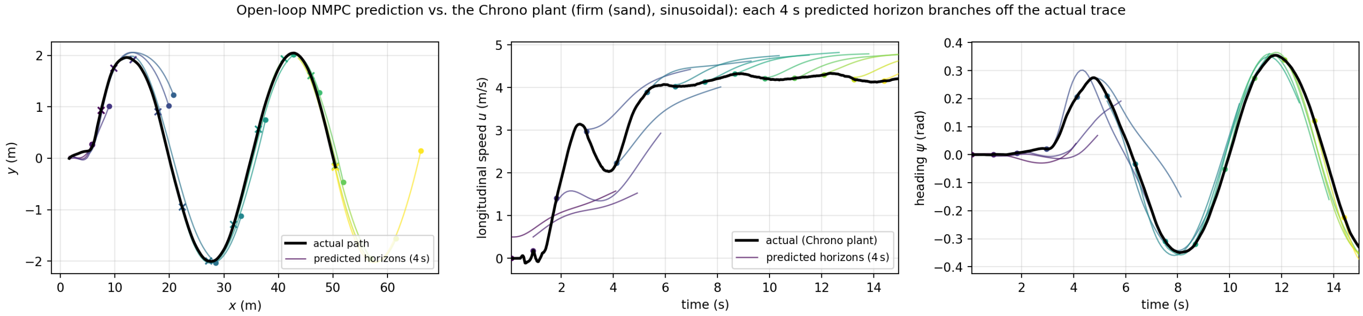

Fig. 6: Open-loop NMPC prediction versus the Chrono plant for a representative firm-sand sinusoidal run (deployed vehicle-rate surrogate). Each 4 s predicted horizon (thin, colour-graded by launch time) branches off the actual plant trace (bold black). **Left (x–y path):** the predicted horizons trace the actual path — the geometry is predicted well — but on firm sand the over-predicted speed carries each predicted endpoint (filled dot) *ahead* of the vehicle's actual position at the same instant (cross): the along-track position drift (≈2.7 m by 4 s). **Middle (speed u):** the horizons run *above* the plant and the gap widens along the horizon — the F_x motion-resistance bias the throttle disturbance observer absorbs online. **Right (heading ψ):** the horizons hug the plant, confirming the lateral (F_y) channel is accurate. Aggregate drifts across the 18-run matrix are quoted in the text (≤ 0.15 rad heading over 4 s; position drift dominated by the longitudinal speed bias, not lateral error).

available input features. Holding the deployed per-axle surrogate architecture fixed (64×32 tanh, scenario split, three seeds), we retrain it with progressively richer inputs — body-frame lateral velocity v, yaw rate ω, road-wheel angle δ, a kinematic lateral-load-transfer estimate, and their first differences — and report held-out RMSE per axle (Table VII). The candidate set is deliberately confined to causal, MPC-available state. Wheel speed is *not* added as a separate input because the slip ratio $\kappa = (\omega_w r - u)/u$ already encodes the wheel-versus-vehicle speed the tire force depends on; promoting wheel speed to a dynamic *state* (rather than an input) is the wheelspeed-state reformulation of [9], out of scope here. Body accelerations (a_x, a_y, a_z) are excluded on principle: by Newton's second law the body accelerations *are* the net forces ($a = \Sigma F/m$), so feeding any acceleration channel to a force predictor is label leakage rather than prediction.

The response is strongly asymmetric. Front-axle lateral force improves by 14% once the dynamic state and its rates are supplied: $F_{y,f}$ genuinely benefits from richer vehicle context. The longitudinal channel is already at its feature floor — F_x RMSE falls at most 3% on either axle — which confirms the Section III-E diagnosis at the model level: the soft-soil speed bias is a dynamics-level motion-resistance term, not a missing tire-model feature, and is therefore exactly the bias the throttle disturbance observer (Section VI) absorbs online rather than something a richer surrogate could learn away.

Rear-axle F_y is a genuine structural floor. Beyond the 4% of Table VII we tested two further physically-motivated hypotheses on dedicated data, and neither moved it. First, per-wheel *sinkage*: because the rear wheels run on soil the front has just compacted, the rear sinks markedly deeper (roughly 50% deeper than the front in a controlled probe), so multi-pass soil state is a plausible missing input; yet adding it improves rear F_y by $\leq 3\%$ *even when the surrogate is handed the ground-truth SCM sinkage as an oracle*. Second, a 0.32 s causal history of the slip and dynamic-state channels (a tire-

TABLE VII: Per-axle held-out force RMSE (N) as causal, MPC-available dynamic features are added to the deployed per-axle rate surrogate (mean over three scenario-split seeds). Front F_y improves 14%; F_x sits at its feature floor ($\leq 3\%$) and rear F_y barely moves (4%), the structural limit characterized in the text.

	F_x (N)		F_y (N)	
Surrogate inputs	front	rear	front	rear
Deployed (rate)	433	579	377	444
+ dynamic state & rates	419	573	**326**	429
Change	−3%	−1%	**−14%**	−4%

relaxation hypothesis) yields 0%. The rear lateral force is thus irreducible from MPC-available signals at the deployed capacity; we treat it as a characterized limit (Section XII-A), not a tuning target.

IV. Online Terrain Estimator

The deployed terrain estimator fuses two complementary measurements of the soil firmness inside a state-augmented UKF (Sec. IV-B): a learned tire-force channel and a proprioceptive vibration channel. This section introduces the proprioceptive channel — a sliding-window MLP — and characterises its closed-loop convergence in isolation; Sec. IV-B then shows why the force channel alone is blind to firm soil and how the covariance-weighted fusion of the two restores observability.

The proprioceptive MLP estimates the Bekker sinkage exponent online from a 4-second window of 26 hand-crafted features over the inertial and wheel-speed streams: speed mean/std/max; lateral and longitudinal acceleration std and high percentiles; wheel-slip mean/std/p95; lateral-grip and yaw-gain proxies; longitudinal drag; mean steering magnitude and p95; throttle mean; and a vertical-dynamics block of vertical-acceleration std/p95, pitch-rate std/p95, and roll-rate std (motivated by Buzhardt & Tallapragada 2024, whose halfcar UKF showed the suspension oscillation signature carries strong Bekker-n information). Every input is available from the simulated onboard sensor stack (IMU, fused body velocity, steering encoder, wheel encoders, commanded throttle); neither SCM tire forces nor true terrain labels are read at inference time.

The MLP channel outputs $\hat{n}$ (and, in the heteroscedastic variant the Fused-UKF consumes, a per-sample uncertainty σ_{MLP}; Sec. IV-B). Each controller tick appends the newest sample to a rolling buffer; once a 4-second window is filled, the feature vector is normalised by the training-set scaler, passed through the MLP, clipped to the trained support, and smoothed by an exponential moving average. The MPC starts from a neutral dirt prior ($n = 0.7$), accepts an estimator update about $4.2\,\mathrm{s}$ into a run, and maps the smoothed $\hat{n}$ onto $(K_\phi, K_c, c, \phi, k)$ by interpolation along the retained clay–dirt–sand terrain manifold. The controller and safety filter therefore receive a complete soil-parameter vector, but the live identified degree of freedom in this paper is n only.

A. Closed-loop convergence

The proprioceptive MLP was trained on a broad multi-axis sweep of 3600 LHS-sampled SCM scenarios spanning $n \in [0.40, 1.30]$ (180 cells), under both scripted open-loop excitation and closed-loop NMPC tracking at three target speeds $(4, 6, 8\,\mathrm{m/s})$ and three reference paths (sinusoidal, lane-change, double-lane-change). With the channel running inside the NMPC at the $10\,\mathrm{Hz}$ control rate, the deployment-mode benchmark drives the same MPC stack across the three canonical soils and six Bekker-jittered out-of-distribution terrains (the five non-n soil parameters perturbed ± 30–$50\,\%$ off the clay–dirt–sand interpolation manifold), each under closed-loop NMPC sinusoidal tracking at $v_{\mathrm{cmd}} \in \{5, 7\}\,\mathrm{m/s}$, bumpiness $\in \{0, 4\}$, five seeds (60 in-distribution and 120 OOD runs).

Table VIII summarises the tail-window error: the proprioceptive channel converges to mean $|\Delta n| = 0.069$ (median 0.066) on the canonical soils and holds 0.081 (median 0.071) across the OOD terrains. Its largest residual is on soft clay, whose very low-K_ϕ response is read slightly upward toward the dirt manifold (tail-mean estimate 0.59 vs. true 0.50, Fig. 7) — the well-known clay/dirt n-ambiguity. This is exactly the regime where the Fused-UKF's lateral-force channel adds complementary information (Sec. IV-B).

a) Out-of-envelope behaviour (bumpiness).: The learned estimator's window features include a vertical-dynamics block (a_z and pitch/roll-rate statistics) that senses how strongly the chassis rides over terrain undulation; these features were trained only at bumpiness $\in \{0, 4\}$ (Table II). At bumpiness 8 — outside that training envelope — the bump-induced vertical excitation is mis-read as firm-soil stiffness and the estimate on soft clay diverges upward: the tail-mean estimate at $n = 0.5$ rises from 0.59 in-envelope to 1.03 ($|\Delta n| \approx 0.53$), i.e. the estimator reports near-firm soil on the softest terrain. This is a genuine out-of-distribution failure on the bumpiness axis, not a low-n failure per se — clay is estimated accurately ($|\Delta n| \approx 0.09$) at bumpiness ≤ 4. It is a limitation of the window-MLP channel alone, not of the deployed Fused-UKF: its state-augmented force channel (Sec. IV-B) infers n through a physics-based bicycle model and does not show this bumpiness sensitivity. Extending the estimator's training set to higher bumpiness, or gating the vertical-dynamics features on an online road-roughness estimate, would close the gap; we report the in-envelope result here and flag the extrapolation regime explicitly rather than averaging over it.

TABLE VIII: Proprioceptive (window-MLP) channel accuracy, evaluated *within its training envelope* (bumpiness $\in \{0, 4\}$; see Sec. III-A and the OOD note below) under closed-loop NMPC sinusoidal tracking ($v_{\mathrm{cmd}} \in \{5, 7\}\,\mathrm{m/s}$, 5 seeds). In-distribution = the three canonical presets (clay/dirt/sand); random soils = Bekker-jittered OOD soils. $|\Delta n|$ is the tail-window absolute error (last $25\,\%$ of the active driving window). This is the proprioceptive channel alone; the deployed Fused-UKF that consumes it is benchmarked in Sec. IV-B.

Set	**Runs**	$\|\Delta n\|$ **mean**	$\|\Delta n\|$ **median**
In-distribution	60	0.069	0.066
Random (OOD) soils	120	0.081	0.071

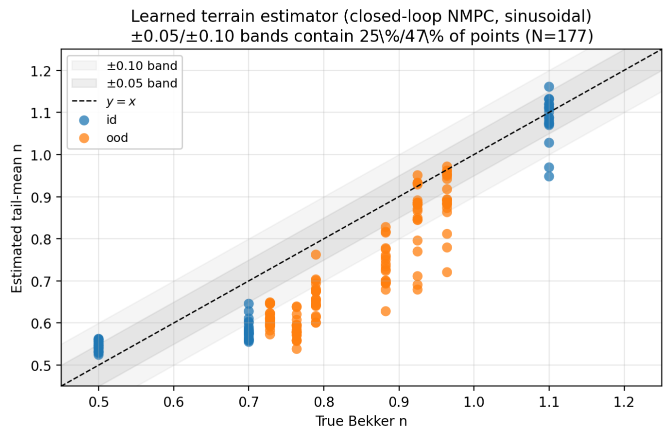

Fig. 7: Learned (sliding-window MLP) terrain estimator: true vs. estimated Bekker n (tail-window mean) on the closed-loop sinusoidal benchmark, evaluated *within the estimator's training envelope* (bumpiness $\in \{0, 4\}$). Blue = in-distribution canonical soils (clay/dirt/sand); orange = Bekker-jittered out-of-distribution soils (n sampled in $[0.40, 1.30]$, other five Bekker parameters perturbed ± 30–$50\,\%$). The ± 0.05 / ± 0.10 bands around $y = x$ contain $40\,\%$ / $68\,\%$ of points. The estimator tracks the diagonal across the deployment range, with a mild systematic over-prediction in the lowest-n (clay) bin (mean estimate 0.59 vs. true 0.50). See the out-of-envelope note in the text for behaviour at bumpiness 8.

B. Closed-loop terrain estimation: force observability and a proprioceptive-fused UKF

The deployed terrain estimator is a **Fused-UKF**: a state-augmented unscented Kalman filter that, following the state-augmentation idea of Dallas et al. [3], estimates the Bekker sinkage exponent n jointly with the bicycle state, but fuses

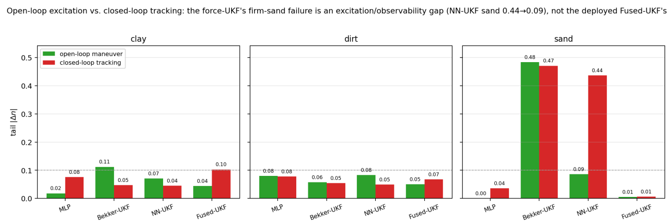

Fig. 8: Open-loop excitation vs. closed-loop tracking, all four estimators, canonical soils (tail $|\Delta n|$). A scripted $0.6\,\mathrm{rad}$ sine-steer maneuver (green) is compared to closed-loop NMPC path-tracking (red). On firm sand the force-only NN-UKF is *observability*-limited — accurate under the scripted maneuver (0.09) but drifting under closed-loop tracking (0.44), so a dedicated maneuver resolves it — whereas the Bekker-UKF remains ≈ 0.47 in both (*model*-limited: its analytical contact law is inaccurate on sand). The proprioceptive MLP is excitation-agnostic ($0.00 \to 0.04$), and the deployed Fused-UKF is accurate in both (≈ 0.01), so it needs no dedicated maneuver at runtime (cf. Sec. IV-B).

two measurement channels of n rather than one. **(i)** A whole-vehicle lateral-force channel: a learned surrogate maps the bicycle's own state plus the six Bekker–Mohr soil parameters to $(F_{y,\mathrm{total}}, M_{\mathrm{yaw,total}})$, and the UKF observes the body-frame a_y as a direct measurement of $\Sigma F_y/m$. **(ii)** A proprioceptive channel: the deployed *heteroscedastic* sliding-window MLP supplies an independent measurement n_{MLP} *together with* its own per-sample uncertainty σ_{MLP}, which the UKF consumes as the proprioceptive measurement noise R_n. The full measurement vector is $[x, y, \psi, u, v, \omega,\, a_y,\, n_{\mathrm{MLP}}]^\top$, and the two n-channels are weighted purely by Kalman covariance — there is *no* hand-tuned gate or external blend. The MLP is trained with a Gaussian negative-log-likelihood objective (val RMSE$_n$ 0.064; $\mathrm{corr}(|\mathrm{err}|, \sigma_{\mathrm{MLP}}) = 0.71$), so it reports low σ exactly where it is accurate — notably on firm sand — and the filter automatically trusts it there.

We compare the deployed Fused-UKF against two force-only baselines that share the identical UKF and bicycle but drop the proprioceptive channel: **(i) Bekker-UKF**, an analytical Bekker–Wong contact-patch integration that mirrors the SCM plant's own formulation, and **(ii) NN-UKF (force-only)**, the whole-vehicle lateral-force surrogate driving the prediction step with no proprioceptive measurement. The bicycle is the 4-wheel double-track variant of Dallas's 3-DOF model — each wheel receives its own vertical load under quasi-static longitudinal and lateral transfer driven by measured a_x, a_y, the steering input, and the body-frame yaw rate.

a) Whole-vehicle lateral-force surrogate.: Earlier iterations trained the UKF's NN on a single-tire rig (paper118 protocol) and wrapped it in a hand-tuned rig-to-vehicle gain ($1.15\times$ on F_y) to close the $\approx 15\,\%$ gap between single-tire rig forces and full-suspension HMMWV forces. The calibration was acceptable at the two soft-soil presets paper118 reported but did not generalise to a broad LHS sweep, so we replaced it with a surrogate that natively predicts what the *whole vehicle* produces:

$$\mathrm{NN}_{F_y} : (u, v, \omega, \delta, K_\phi, K_c, n, c, \phi, k) \;\mapsto\; (F_{y,\mathrm{total}},\; M_{\mathrm{yaw,total}}).$$

The surrogate is a 128–64 ReLU MLP trained on a 300-sample, 9-axis uniform LHS sweep (seed 7) that varies the six Bekker–Mohr soil parameters together with three maneuver axes — steering amplitude 0.20–$0.65\,\mathrm{rad}$, steering period 2–5 s, and target cruise speed 3.5–$7\,\mathrm{m/s}$ (half the runs use constant open-loop throttle, half a PI cruise) — yielding $\approx 190\,000$ labelled rows. Crucially, the soil axes are sampled from a *widened* box (K_ϕ from $0.3\,\mathrm{MPa}$, cohesion from $300\,\mathrm{Pa}$, Janosi shear 0.008–$0.028\,\mathrm{m}$, $n \in [0.35, 1.35]$): the canonical clay, dirt and sand presets sit on the *edges* of the deployed `TRAINING_RANGES_V6` box (clay's K_ϕ is at $5\,\%$ of range and its Janosi shear at the $0\,\%$ floor), so a surrogate trained on that box has to extrapolate to reach them and fails on clay. Widening the box (never narrowing, so the uniform LHS coverage is preserved) turns those soils into interior points the surrogate interpolates to. Targets are extracted directly from each Chrono log: $F_{y,\mathrm{total}} = m\,a_y$ and $M_{\mathrm{yaw,total}} = I_z\,\dot{\omega}$ (5-point central difference). Held-out (30 of 300 scenarios, scenario-split) R^2 is 0.90 on F_y and 0.88 on M_{yaw}. The sampling discipline is strict: every input axis is uniformly sampled over its (widened) box, with no narrowing toward a canonical preset or controller-conditioned operating point, and the trained model is plugged into the UKF *without* any per-soil scalar correction. This surrogate is the UKF's force channel for both the Fused-UKF and the force-only NN-UKF.

b) Force observability of n (the key issue).: The reason the force channel alone is not enough is an *observability* gap, not a model defect. Under a path-tracking controller the slip stays small, and at that operating point the lateral force carries almost no information about n on firm soil. At a representative small-slip operating point the lateral-force signal separating firm from soft soil is $|F_y(n{=}1.1) - F_y(n{=}0.7)| \approx 132\,\mathrm{N}$, well below the lateral-accel measurement-noise floor of $\approx 770\,\mathrm{N}$ ($= m\,R_{a_y}$ with $R_{a_y} = 0.3\,\mathrm{m/s^2}$). Under a *scripted* $0.6\,\mathrm{rad}$ steering maneuver the same delta grows to $743\,\mathrm{N}$ — at the noise floor — and $\mathrm{d}F_y/\mathrm{d}n$ is $\approx 16\times$ larger. The yaw channel is worse still: $|\mathrm{d}M_z|\,\Delta n \approx 191\,\mathrm{N{\cdot}m}$ sits below its $\approx 216\,\mathrm{N{\cdot}m}$ noise floor and stays unobservable even at large slip. A force-only UKF is therefore *blind* to firm-soil n under gentle tracking: the signal it would need to correct n is buried in the measurement noise. Fig. 9 makes this concrete — and shows that the proprioceptive channel, which reads vibration rather than slip-dependent force, fills exactly this gap.

c) Open-loop vs. closed-loop: two distinct failure modes.: The earlier open-loop diagnostic (Fig. 8) separates an *observability* limit from a *model* limit by comparing each estimator's tail $|\Delta n|$ on firm sand under a scripted excitation maneuver against the same soil under closed-loop tracking. The force-only NN-UKF goes from 0.09 (open-loop) to 0.44 (closed-loop): it is observability-limited, and a dedicated excitation maneuver *resolves* it. The Bekker-UKF, by contrast, is essentially unchanged across the two ($0.48 \to 0.47$): it is *model*-limited — the analytical Bekker law is inaccurate on sand, and

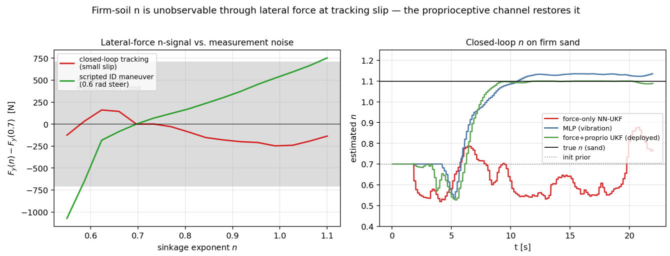

Fig. 9: Force observability of the sinkage exponent n. **Left:** the lateral-force n-signal $|F_y(n=1.1) - F_y(n=0.7)|$ against the lateral-accel measurement-noise band ($m\,R_{a_y} \approx 770\,\mathrm{N}$, $R_{a_y} = 0.3\,\mathrm{m/s^2}$). At closed-loop small-slip path tracking the signal ($\approx 132\,\mathrm{N}$) stays *inside* the band — n is unobservable from force — whereas a scripted $0.6\,\mathrm{rad}$ maneuver escapes it ($\approx 743\,\mathrm{N}$, with $\mathrm{d}F_y/\mathrm{d}n \approx 16\times$ larger). **Right:** closed-loop $\hat{n}(t)$ on firm sand: the force-only NN-UKF, lacking an observable force signal, drifts in the *wrong* direction, while the deployed Fused-UKF tracks the true n by leaning on its proprioceptive channel.

no amount of excitation can compensate. The proprioceptive MLP, which reads vibration and is excitation-agnostic, is accurate in both ($0.00 \to 0.04$), and the Fused-UKF inherits that robustness ($0.01 \to 0.01$). This is the observability story seen from the other side, and it is why the deployed estimator needs *no* dedicated excitation maneuver at runtime.

d) The deployed Fused-UKF closes the gap (canonical presets).: Fig. 10 reports the four estimators run *live in the closed loop* on the canonical clay/dirt/sand presets at $v \in \{5, 7\}\,\mathrm{m/s}$ (three seeds each); all numbers are tail $|\Delta n|$ (lower is better). The proprioceptive MLP gives clay 0.076, dirt 0.078, sand 0.036 (overall 0.063); the Bekker-UKF 0.047, 0.054, 0.471 (overall 0.190); the force-only NN-UKF 0.045, 0.049, 0.437 (overall 0.162); and the deployed Fused-UKF 0.103, 0.068, 0.006 (overall 0.059). The two force-only UKFs collapse on firm sand (≈ 0.44–0.47) for exactly the observability reason above, whereas the proprioceptive channel makes n observable, so the Fused-UKF accurately recovers firm sand ($|\Delta n| = 0.006$) and is the most accurate estimator overall (0.059). We note one limitation plainly: the Fused-UKF is the *least* accurate on canonical clay (0.103) — both force-only UKFs and the MLP outperform it there, because the proprioceptive channel is slightly over-weighted on soft, on-manifold soil where the force channel is already reliable. The trade is nonetheless favourable: the deployed filter accepts a small clay penalty in exchange for eliminating the firm-sand failure exhibited by both force-only baselines.

e) Broad robustness across the deployable n range.: Fig. 12 and Table IX report the head-to-head on **100 soils swept along the canonical clay–dirt–sand Bekker–Mohr manifold** (seed 42, $n \in [0.52, 1.08]$; the other five soil parameters follow the preset manifold with $\pm 8\,\%$ jitter — the parameter-known setting in which the force-dominant exponent n is isolated, and the soil model the estimator reconstructs from $\hat{n}$), *all* run live closed-loop under NMPC sinusoidal path-tracking at $5\,\mathrm{m/s}$. The deployed Fused-UKF and the proprioceptive MLP recover n across the full range (median error $4.5\,\%$ and $6.7\,\%$; $94\,\%$ and $83\,\%$ within $\pm 20\,\%$), their estimates tracking the diagonal. The force-only NN-UKF and Bekker-UKF instead saturate near $n \approx 0.6$ on firm soils ($23.8\,\%$ median error): without a proprioceptive channel they cannot observe n once closed-loop slip is small — the force-observability limit of Sec. IV-B, now visible across the whole sweep rather than only on canonical sand. The deployed Fused-UKF leads every metric (Fig. 13).

TABLE IX: Closed-loop terrain-estimator head-to-head on 100 soils swept along the clay–dirt–sand Bekker–Mohr manifold ($n \in [0.52, 1.08]$, seed 42), all run live under closed-loop NMPC sinusoidal path-tracking at $5\,\mathrm{m/s}$. The other five soil parameters follow the preset manifold ($\pm 8\,\%$ jitter), the parameter-known setting for isolating the force-dominant exponent n. Errors are tail-window $|\Delta n|$ (absolute) and $|\Delta n|/n_{\mathrm{true}}$ (relative). The deployed Fused-UKF row is bold.

Estimator	median $\lvert\Delta n\rvert$	median %err	within $\pm 20\,\%$
MLP	0.055	6.7 %	83 %
Bekker-UKF	0.189	23.8 %	41 %
NN-UKF	0.186	23.8 %	44 %
Fused-UKF	**0.028**	**4.5 %**	**94 %**

V. Terrain-Aware Speed Planning

The reference-tracking NMPC follows a speed reference $v_{\mathrm{ref}}(s)$ along the path. A purely geometric curvature-limited profile ($v_{\mathrm{ref}} = \sqrt{a_{y,\max}/|\kappa(s)|}$ at a fixed $a_{y,\max}$) is blind to the soil: on soft terrain it commands cornering speeds the available grip cannot support, so the vehicle understeers off the line. The failure is most visible on the high-speed sinusoid, exactly where the analytical baselines of Sec. III-B pile up (Fig. 2).

We replace it with a terrain-aware, quasi-steady-state time-optimal profile over a friction-circle (g–g) budget. At each control step the deployed tire surrogate is queried at the live estimate $\hat{n}$ and the per-axle vertical load for the available lateral force ($\to a_{y,\max}$) and tractive/braking force ($\to a_{x,\max}$, $a_{x,\mathrm{brake}}$); a forward–backward pass then returns the fastest $v_{\mathrm{ref}}(s)$ that keeps the combined demand inside the circle,

$$\left(\frac{a_x}{a_{x,\max}}\right)^2 + \left(\frac{v^2\,\kappa(s)}{a_{y,\max}}\right)^2 \leq 1, \qquad (1)$$

with a grip-safety margin that de-rates the surrogate's peak-force query to leave transient headroom. The curvature $\kappa(s)$ is read from the analytic arc-length spline of the path rather than finite-differenced over the horizon, whose node spacing varies with speed and otherwise produces spurious κ spikes that collapse v_{ref}. This is a third use of the differentiable surrogate, alongside in-horizon force prediction and the traction-budget constraint: the same model used for tracking also determines the cornering speed the soil can support.

A closed-loop A/B against the geometric reference (clay/dirt/sand $\times$ {sinusoid, double lane change} $\times \{5, 7\}\,\mathrm{m/s}$, five seeds; Fig. 14) cuts mean RMS CTE by

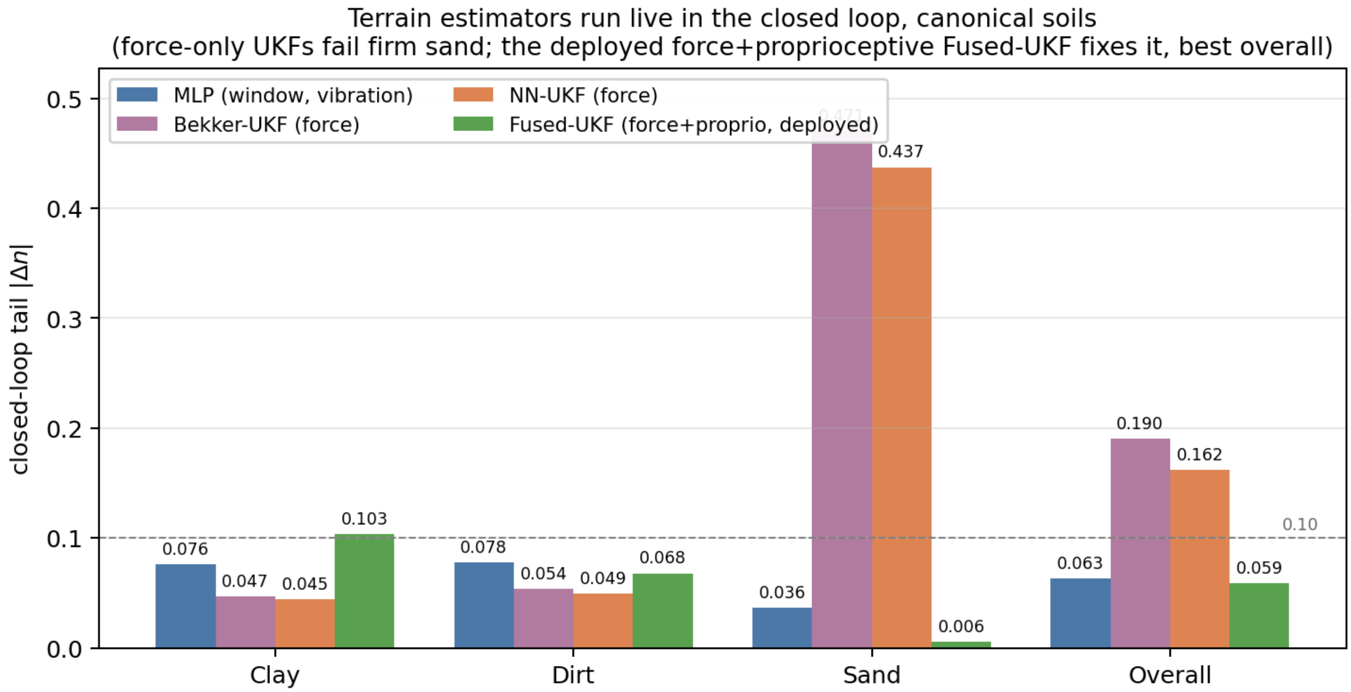

Fig. 10: Per-terrain closed-loop comparison of the four estimators on the canonical Chrono SCM presets — clay, sandy loam (dirt), dry sand — run *live* in the loop at $v \in \{5, 7\}$ m/s, three seeds, 0.6 rad steering, tail $|\Delta n|$. The Bekker-UKF and force-only NN-UKF are most accurate on the soft soils but fail on firm sand ($|\Delta n| \approx 0.44$–0.47, an observability limit, cf. Fig. 9); the proprioceptive MLP is balanced; the deployed Fused-UKF recovers firm sand to 0.006 at the cost of being least accurate on soft clay (0.103).

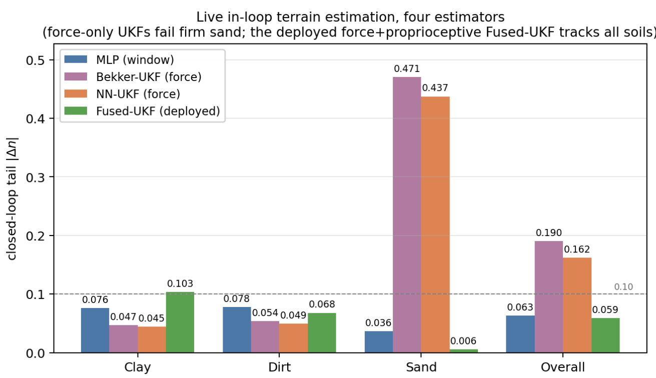

Fig. 11: Deployment summary: tail $|\Delta n|$ over the canonical clay/dirt/sand presets and the overall mean for the proprioceptive MLP, the analytical Bekker-UKF, the force-only NN-UKF, and the deployed Fused-UKF (all run live in the closed loop, $v \in \{5, 7\}$ m/s, three seeds). The force-only UKFs fail firm sand; the covariance-fused proprioceptive channel makes n observable, so the Fused-UKF is best overall (0.059) with no hand-tuned gate.

51 % ($0.156 \rightarrow 0.077$ m) for a 6 % reduction in mean speed, and collapses the across-scenario spread (std $0.19 \rightarrow 0.03$ m). Here RMS CTE is read as a *controllability* outcome — whether the vehicle can hold the planned line under the available grip, rather than a path-tracking accuracy target — since a profile that ignores the soil commands cornering speeds that slide the vehicle off the line. The gain concentrates on exactly the cells where the curvature heuristic commands cornering speeds the soil cannot support — the high-speed sinusoid on soft soil, where CTE falls by 70–84 % (dirt $0.561 \rightarrow 0.168$, clay $0.567 \rightarrow 0.090$ m) — and is neutral on the already-benign cells. The trade-off is grip-limited rather than tunable: raising the safety margin to recover the lost speed yields little additional speed (on dirt at 7 m/s, +0.3 m/s) while tripling CTE ($0.10 \rightarrow 0.28$ m), so the chosen operating point is near the best achievable speed/accuracy trade-off. Because the profile reads its grip limits from whichever tire model the controller carries, the tire-model comparison of Sec. III-B holds it fixed at the geometric reference to isolate the tracking model; the system-level closed-loop sweeps elsewhere in the paper run the deployed g–g profile.

VI. Asymmetric Throttle Disturbance Observer

The bicycle-model NMPC integrates the longitudinal velocity u under the open-loop dynamics $\dot{u} = a_x$, with a_x the commanded longitudinal acceleration. On deformable soil the realised acceleration is consistently smaller than commanded because of sinkage drag, producing a steady-state under-speed bias. A first-order integral disturbance observer on the actuation map absorbs this bias:

$$\dot{d} = \begin{cases} K_i(u_{\text{ref}} - u), & u < u_{\text{ref}}, \\ -K_b\, d, & u \geq u_{\text{ref}}, \end{cases} \qquad |d| \leq d_{\max}, \qquad (2)$$

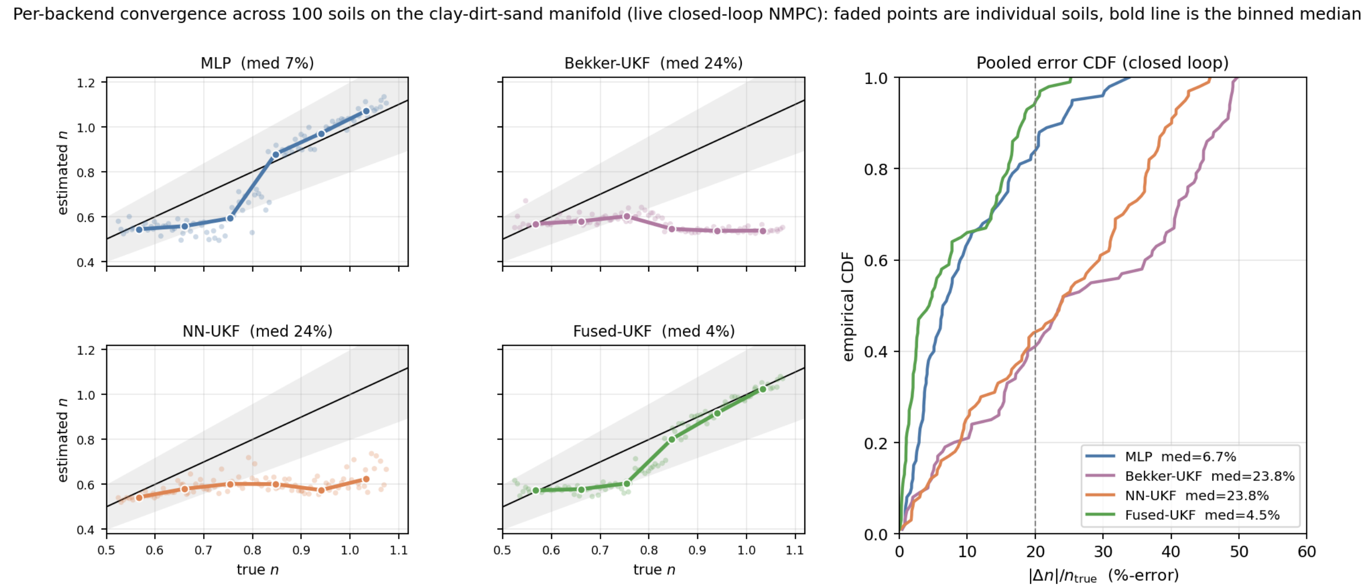

Fig. 12: Closed-loop terrain-estimator benchmark on 100 soils swept along the clay–dirt–sand Bekker–Mohr manifold in Chrono SCM (seed 42, $n \in [0.52, 1.08]$; the other five soil parameters follow the preset manifold with $\pm 8\,\%$ jitter — the parameter-known setting in which the force-dominant exponent n is estimated, and the soil model our estimator reconstructs from $\hat{n}$), all run live under closed-loop NMPC sinusoidal path-tracking at $5\,\text{m/s}$. Left, four panels: estimated n versus true n per backend, where faded points are individual soils, the bold line is the binned median, and the black diagonal with its $\pm 20\,\%$ band is the target. The proprioceptive MLP and the deployed Fused-UKF track the diagonal (fitted slope 1.25 and 1.07), whereas the force-only NN-UKF and Bekker-UKF saturate near $n \approx 0.6$ on firm soils (slope 0.15 and -0.11) — the closed-loop force-observability limit quantified in Sec. IV-B. Right: pooled empirical CDF of the tail-window $|\Delta n|/n_{\text{true}}$; the Fused-UKF holds the tightest distribution and the best fraction within $\pm 20\,\%$ (dashed line).

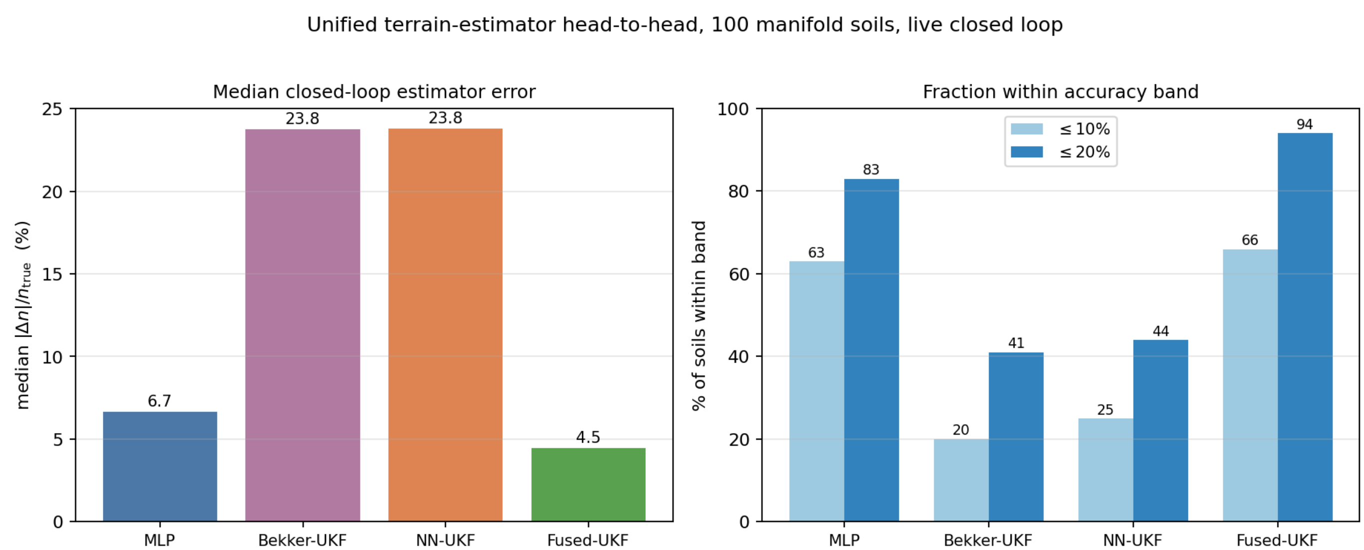

Fig. 13: Pooled closed-loop terrain-estimator comparison over the 100 manifold soils. **Left:** median $|\Delta n|/n_{\text{true}}$. **Right:** fraction within the $\pm 10\,\%$ / $\pm 20\,\%$ bands. The deployed Fused-UKF is the most accurate ($94\,\%$ within $\pm 20\,\%$) and the MLP second ($83\,\%$); the force-only NN-UKF and Bekker-UKF trail because they cannot observe n on firm soils in closed loop, with no hand-tuned gate needed in the Fused-UKF.

with the bleed gain $K_b < K_i$ to make the observer asymmetric: it accumulates rapidly during under-speed and decays slowly toward zero, never contributing a positive over-speed bias. The applied throttle is $\tau = \tau_{\text{NMPC}} + d$. The ablation zeros both gains ($K_i = 0$, $d_{\max} = 0$), keeping the structural hook but suppressing all integral action so the comparison is apples-to-apples. Table X and Fig. 15 report the result over 405 closed-loop runs per variant.

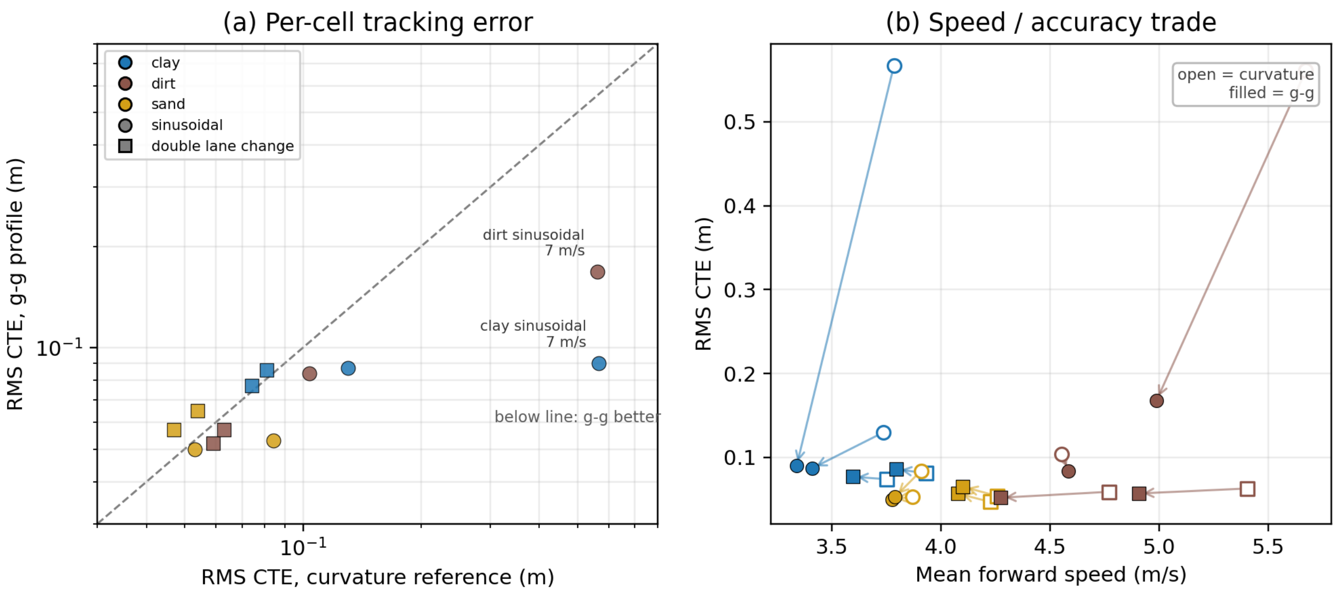

Fig. 14: Terrain-aware g–g speed profile vs. the geometric curvature reference, closed loop. **(a)** Per-cell RMS CTE: the two soft-soil high-speed cells (far right) drop to the trackable cluster, while benign cells are unchanged (a few sand cells sit marginally above the diagonal, the profile being slightly conservative there). **(b)** Speed/accuracy trade: open markers are the curvature reference, filled the g–g profile; arrows show each cell trading a little mean speed for a large CTE reduction. The grip limits come live from the tire surrogate at $\hat{n}$ (Sec. III).

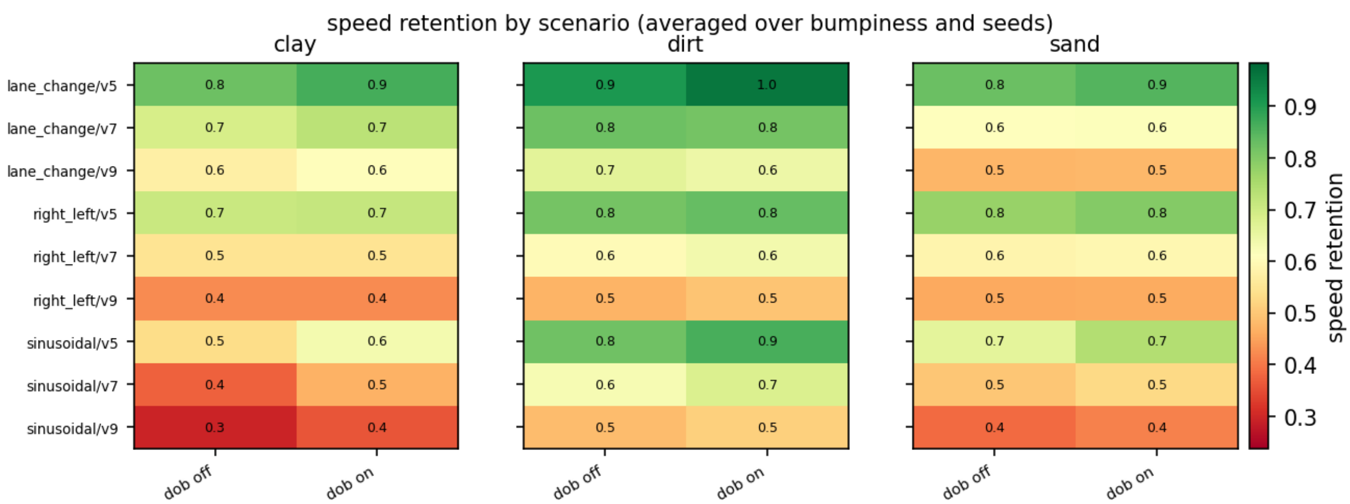

Fig. 15: Throttle disturbance observer ablation: per-scenario speed retention (achieved $\bar{u}$ divided by reference v_{ref}) heatmap; row labels show v_{ref} and the achieved $\bar{u}$ so the gap between target and realised speed is explicit. The observer's gain is largest on sand at $v_{\mathrm{ref}} = 7\,\mathrm{m/s}$, consistent with the higher sinkage drag in that regime.

A. Why a reactive observer rather than a feedforward model

A natural alternative is to make the resistance *feedforward*: the OCP already exposes a longitudinal residual ($\dot{u} = a_x + du_{\mathrm{resid}}$), so one can set $du_{\mathrm{resid}} = -c_{\mathrm{drag}}(\hat{n})$ with c_{drag} calibrated from the DOB-off rollout drift and indexed by the live estimate $\hat{n}$. We implemented and calibrated this, and it clarifies why the reactive observer is the better choice. First, the prediction bias is *not* a monotonic drag: only firm soil (dry sand) genuinely over-predicts speed ($c_{\mathrm{drag}} \approx 0.17\,\mathrm{m/s^2}$); on dirt the kinematic prediction *under*-predicts and on clay it is accurate, so the calibrated feedforward is effectively sand-specific. Second, a head-to-head over clay/dirt/sand (Table XI) shows the feedforward term gives only a marginal gain on the calibrated firm sand (sand speed ratio $0.60 \to 0.61$), is inert on clay, and is mildly adverse on dirt ($0.77 \to 0.76$, which under-predicts rather than over-predicts), but it *cannot replace* the observer: the DOB is higher on every terrain (clay 0.64 vs. 0.51, sand 0.65 vs. 0.61) because clay's deficit is a

TABLE X: Asymmetric throttle disturbance observer ablation. 405 runs per variant. Speed ratio is reported against the *terrain-aware g-g speed profile* (Sec. V; mean $u_{\text{meas}}/\text{mean}\,v_{\text{ref}}(s)$) rather than the CLI cruise target, because on curvature-heavy paths the profile itself caps below the cruise target – the cruise target is the *intent*, the profile is what the controller is asked to track. The observer increases achieved speed by approximately $5\,\%$ at a small ($\sim$7 mm) increase in mean tracking error.

Variant	RMS CTE (m)	Speed/profile	$\bar{u}$ (m/s)
DOB off	0.094 ± 0.043	0.61	4.07
DOB on	0.101 ± 0.088	**0.64**	**4.26**

TABLE XI: Feedforward sinkage-drag vs. the reactive DOB — mean speed ratio (achieved $\bar{u}$ / reference), sinusoidal path, $v \in \{5, 7\}$ m/s, two seeds. *off*: neither; *ffdrag*: feedforward drag, DOB off; *DOB*: reactive observer. The feedforward term helps where calibrated (sand, dirt) but trails the DOB on every terrain.

Terrain	off	ffdrag	DOB
clay	0.50	0.51	**0.64**
dirt	0.77	0.76	**0.80**
sand	0.60	0.61	**0.65**

traction limit, not a drag, which only integral feedback closes. Combining the two does not raise speed and inflates tracking error. A terrain-dependent, sign-varying bias is exactly the regime where an adaptive observer beats a fixed model, so the DOB is retained as the default and the feedforward term is left as an optional firm-soil refinement.

The same conclusion holds for a feedforward at the *actuation map* itself, the most direct DOB replacement. The naive map $\tau = a_x/a_{x,\max}$ omits a terrain-dependent throttle offset; the integral DOB converges to exactly that offset, which is smooth in $\hat{n}$ (clay/dirt $\approx$0.24, sand $\approx$0.07). Baking the converged offset in as a static held bias $d_{\text{ff}}(\hat{n})$ and removing the integral (a 48-run clay/dirt/sand ablation) again fails to replace the observer: it recovers part of the clay deficit ($2.98 \to 3.13$ vs. DOB 3.41 m/s) but *over*-throttles firmer dirt ($4.55 \to 4.43$, below even no compensation) and raises tracking error, because the required offset is not a static function of $\hat{n}$ alone — it varies with instantaneous speed, curvature, and load transfer, which the integral tracks online and a terrain-indexed map cannot. Stacking feedforward and integral matches the DOB's speed but inflates clay CTE ($0.06 \to 0.11$ m). The longitudinal disturbance-rejection burden is therefore genuinely adaptive: the DOB is not a tunable patch over a poor model but a necessary closed-loop element.

VII. Two-Layer Obstacle Avoidance

The planner adds a smooth softplus barrier to the stage and terminal costs for each known obstacle. Downstream of the planner and human command source, the `SAFETY_FILTERS` registry exposes a common `filter` interface: the filter receives the desired steering, throttle, brake, the current state, and the nearby obstacles, and returns the filtered command plus diagnostics. The DOB-CBF variant solves a single-step minimum-deviation QP around the incoming command — a minimally-invasive control-barrier-function filter [12] in the shared-control tradition of predictive and shared-autonomy teleoperation [14], [15], here with a disturbance-observer traction budget and physical rate limits; it is the natural intent-preserving mechanism for HIL teleoperation — and logs intervention rate and command deviation as an intrusiveness signal. It also carries a rear-approach guard: when an in-lane vehicle is closing from behind within a threat distance, it refuses any deceleration and, with the path ahead clear, accelerates to escape (even from a standstill), since slowing into a rear approach only worsens the impact (scope and limits in Sec. XII-A). The DOB-CBF output is passed through a steering/throttle actuator model — a physical rate limit imposed in the QP and enforced at the physics step — so the filter never commands a non-physical instantaneous reversal that would otherwise destabilise the front end under teleoperation.

a) Comparison.: The safety layer is evaluated as DOB-CBF (the intent-preserving filter of Zhang et al. [8], optionally reading the same neural surrogate) versus no filter. The registry remains swappable, but DOB-CBF is the default instance: it is the only candidate that screens a human command by minimum deviation rather than re-optimising it, which is what teleoperation requires.

A. Human-in-the-loop safety-filter protocol

The safety filter exists to screen a *human operator's* delayed commands; the framework therefore includes a human-in-the-loop (HIL) protocol in addition to the autonomous sweeps reported below. A human drives the HMMWV through the rock field from the Chrono::Sensor driver point-of-view using a Logitech G29 steering wheel, while the sim-side filter screens the operator commands immediately before vehicle actuation. Each round delays *both* teleoperation channels: the operator command path (uplink) by $\Delta_{\text{cmd}} \in \{0, 0.15, 0.30\}$ s and the driver-camera feed (downlink) by $\Delta_{\text{cam}} = s\,\Delta_{\text{cmd}}$, where the camera-delay scale s defaults to 1.0 for a symmetric link. The same script supports a profile-driven delay for learned 5G traces.

The HIL layer reuses the same command contract, swappable safety filters, latency injection, and metric pipeline as the autonomous evaluation, logging the same safety and intrusiveness signals (collisions, minimum clearance, intervention rate, mean steering and throttle/brake correction, speed retention). Fig. 16 shows the operator's view: the Chrono::Sensor driver point-of-view over the deformable terrain and boulder field, with the live heads-up display overlaying the delayed command and the filtered command actually sent to the vehicle.

a) Scope.: The HIL pipeline is included to show the framework can drive a real human operator end-to-end — G29 input, Chrono::Sensor camera POV, the live HUD, and per-channel latency, all through the same command contract and safety filter as the autonomous stack (Fig. 16). This

Fig. 16: The human-in-the-loop operator view: the Chrono::Sensor driver point-of-view over the SCM deformable terrain and boulder field (the same camera feed that carries the downlink latency), with the live HUD (`hil_hud.py`) overlaying the steering and throttle/brake state — the ghost mark is the delayed operator command, the solid mark the filtered command sent to the vehicle. The pipeline is a demonstrated capability; the quantitative safety result is obtained from the deterministic counterfactual of Sec. VII-F, not from human rounds.

is a systems and control paper, not a human-factors study, so the *quantitative* safety evaluation is deliberately not a human-operator comparison (which would carry inter-operator variance and a separate experimental protocol). We instead validate the filter with the deterministic *counterfactual* of Sec. VII-F: replaying a fixed command trace (filter-off versus DOB-CBF-on) over the *identical* scenario, so the collision-prevention measurement is paired and reproducible. The traces used are worst-case "drive-into-the-hazard" commands that stress the filter; because the simulator is deterministic, the same replay machinery can also be applied to recorded G29 traces without changing the evaluation code.

B. Autonomous validation: safety-filter-only collision avoidance

The HIL protocol of Section VII-A is complemented by an autonomous controlled experiment in which an obstacle-blind planner stands in for the operator, isolating the safety filter as the sole collision avoider. This removes operator variance and supports a larger, fully repeatable matrix (405 trials per filter; 810 trials total). The same matrix is re-evaluated with the planner's in-horizon barriers re-enabled in Sec. VII-C for a total of 1620 runs. Table XII and Fig. 17 report the safety-filter-only outcome. Because the planner's reference here is deliberately routed through the obstacles, crosstrack error would conflate the beneficial avoidance manoeuvre with poor tracking and is therefore not reported; intrusiveness is instead measured *directly*, by how often a filter overrides the planner (intervention rate) and how hard it corrects the command (mean absolute steering and throttle correction, $|\Delta\mathrm{steer}|$, $|\Delta\mathrm{thr}|$) — the intent-preservation cost a teleoperator would feel. DOB-CBF cuts mean collisions by $79\,\%$ ($3.07 \rightarrow 0.66$ per run) and turns a mean clearance of $-1.81\,\mathrm{m}$ (inside the obstacle footprint) into $+0.50\,\mathrm{m}$, while intervening on $57\,\%$ of steps with small corrections ($|\Delta\delta| = 0.28$, $|\Delta\tau| = 0.39$) — the minimum-deviation, intent-preserving behaviour that motivates it as the default safety filter.

TABLE XII: Autonomous safety-filter-only validation (planner blind to obstacles). 405 runs per filter (810 total); same scenario matrix as Table XIII, blind rows only. DOB-CBF, the default intent-preserving filter (Sec. VII-A), cuts collisions $79\,\%$ and recovers a positive clearance margin at a low intervention rate. Collisions and clearance are mean±std over the 405 runs/filter; $|\Delta\delta|$, $|\Delta\tau|$ are the mean absolute steering and throttle command corrections the filter applies.

Filter	**Coll./run**	**Clearance (m)**	**Interv. (%)**	$\boldsymbol{|\Delta\delta|}$	$\boldsymbol{|\Delta\tau|}$
None	$3.07{\pm}1.18$	$-1.81{\pm}0.72$	—	—	—
DOB-CBF	$\mathbf{0.66{\pm}0.76}$	$\mathbf{+0.50{\pm}1.97}$	57	0.28	0.39

TABLE XIII: Planner-aware vs. planner-blind safety sweep. 1620 runs. The two-layer stack (barrier + DOB-CBF) drives collisions to 0.13/run while *reducing* intervention and command deviation relative to the filter-only (blind) configuration.

Variant	**Coll./run**	**Clearance (m)**	**Interv. (%)**	$\boldsymbol{|\Delta\delta|}$	$\boldsymbol{|\Delta\tau|}$
None, blind	$2.99{\pm}1.22$	$-1.81{\pm}0.36$	—	—	—
None, aware	$0.52{\pm}0.87$	$+0.41{\pm}1.08$	—	—	—
DOB-CBF, blind	$0.66{\pm}0.75$	$+0.40{\pm}2.03$	56	0.29	0.36
DOB-CBF, aware	$\mathbf{0.13{\pm}0.44}$	$\mathbf{+2.00{\pm}1.64}$	40	0.14	0.31

C. Two-layer (planner-aware) stack

With the in-horizon NMPC barrier re-enabled the two-layer stack is evaluated. Table XIII reports 1620 runs total across the two safety-filter variants (none, DOB-CBF) under both planner-aware and planner-blind modes. Barriers alone (no-filter, planner-aware) cut collisions by approximately $6\times$ relative to the planner-blind baseline ($2.99 \rightarrow 0.52$ per run), and combining the barrier with the DOB-CBF filter drives collisions to 0.13 per run — a $96\,\%$ reduction over the blind baseline — at the lowest intervention rate and command deviation of any configuration, which is exactly the two-layer, intent-preserving behaviour the stack is designed for.

D. DOB-CBF neural-surrogate ablation

The DOB-CBF filter can run either with the neural tire surrogate enabled or with a kinematic/linear fallback. In the fallback mode the disturbance observer uses fixed longitudinal acceleration limits, the CBF uses kinematic steering authority, and the terrain speed cap omits the NN traction check. With the NN enabled, the same QP receives terrain-aware longitudinal acceleration, lateral-force, yaw-moment, and traction-limit estimates from the whole-vehicle surrogate. Table XIV and Fig. 18 compare the two variants over 1215 closed-loop trials.

E. Planner tire model under a fixed DOB-CBF filter

A separate sweep runs the autonomous obstacle scenario with the DOB-CBF filter always on, varying only the planner's tire model (the analytical planners use the SCM-calibrated parameters of Sec. III-B). All three tire models avoid the

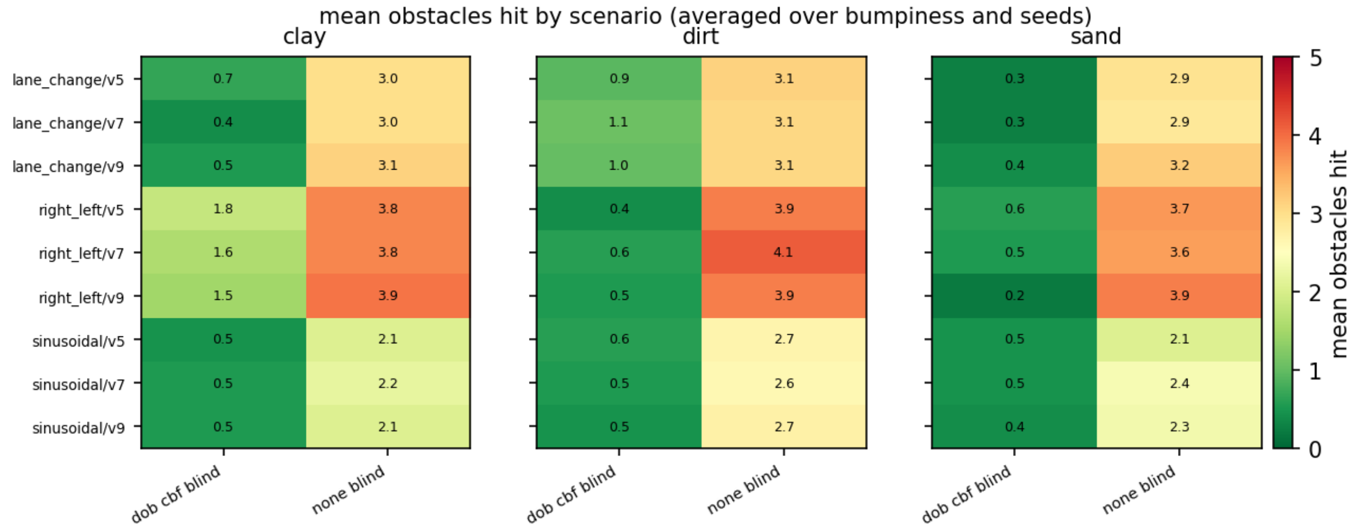

Fig. 17: Autonomous safety-filter-only validation. Mean unique obstacles hit per run, stratified by terrain and safety filter, with whisker error bars showing across-seed standard deviation. The reported sweep covers all three deployment terrains (clay, dirt, sand) across three speeds and three bumpiness levels, 135 runs per (filter, terrain) cell. Aggregate KPIs are in Table XII.

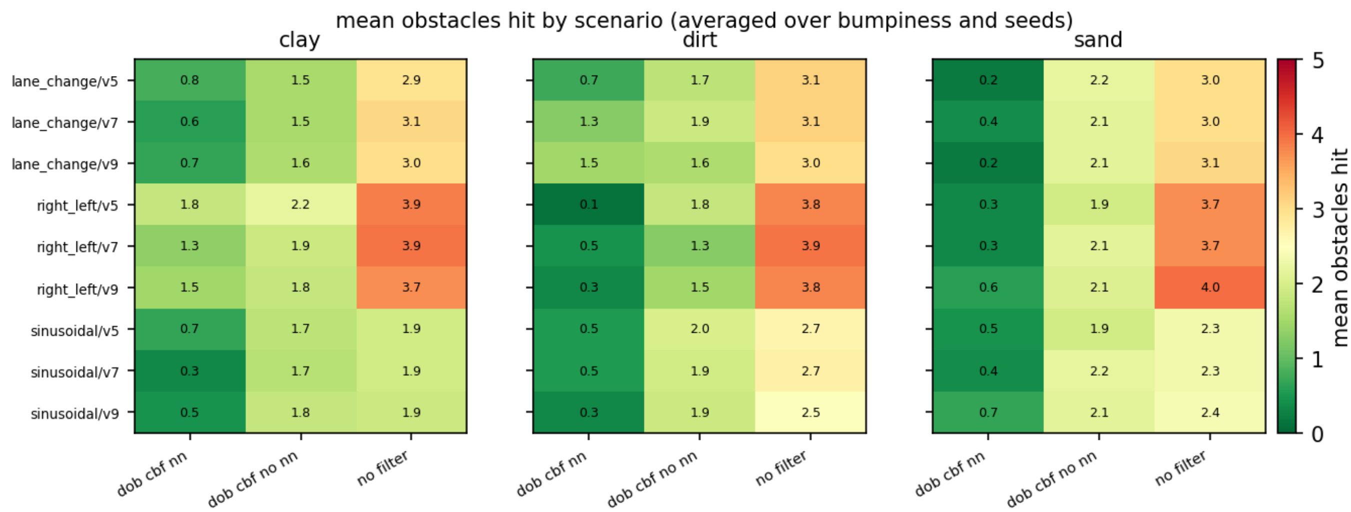

Fig. 18: DOB-CBF neural-surrogate ablation: per-scenario collisions. Aggregate KPIs are in Table XIV.

TABLE XIV: DOB-CBF neural-surrogate ablation. 1215 runs.

Variant	**Coll./run**	**Clearance (m)**	**Interv. (%)**	$\lvert\boldsymbol{\Delta\delta}\rvert$	$\lvert\boldsymbol{\Delta\tau}\rvert$
None (no filter)	$3.06{\pm}1.20$	$-1.84{\pm}0.32$	—	—	—
DOB-CBF, no NN	$1.86{\pm}0.74$	$-0.62{\pm}0.38$	44	0.29	0.27
DOB-CBF, NN	$\mathbf{0.65{\pm}0.78}$	$\mathbf{+0.32{\pm}1.76}$	56	0.28	0.36

TABLE XV: Autonomous obstacle avoidance, planner tire model swept under a fixed DOB-CBF filter (analytical planners SCM-calibrated). 1215 runs (405 per model).

Planner tire model	**Coll./run**	**Clearance (m)**	**CTE (m)**
Pacejka	$0.64{\pm}0.95$	$+0.81{\pm}1.99$	0.77
TMeasy	$0.67{\pm}1.00$	$+0.77{\pm}2.00$	0.84
NN rate-MLP	$0.61{\pm}0.98$	$+0.73{\pm}1.91$	0.90

obstacles approximately equally (Table XV and Fig. 19), indicating that the safety filter dominates the filter-mediated outcome and that the planner-tire choice is largely orthogonal to the safety outcome.

F. Counterfactual replay on dynamic-traffic (convoy) scenarios

The evaluations above use static rock obstacles. To stress the filters against *dynamic* hazards we add PID-driven traffic vehicles — full HMMWVs sharing the ego's deformable-terrain system and collision world, each following a lane at a set speed with scripted dangerous behaviour (a lead vehicle braking, a cut-in, a vehicle stalled in the lane). They enter the same obstacle interface as the rocks, so the ego planner and the safety filter must avoid them with no change to either.

These scenarios enable the *counterfactual* measurement that

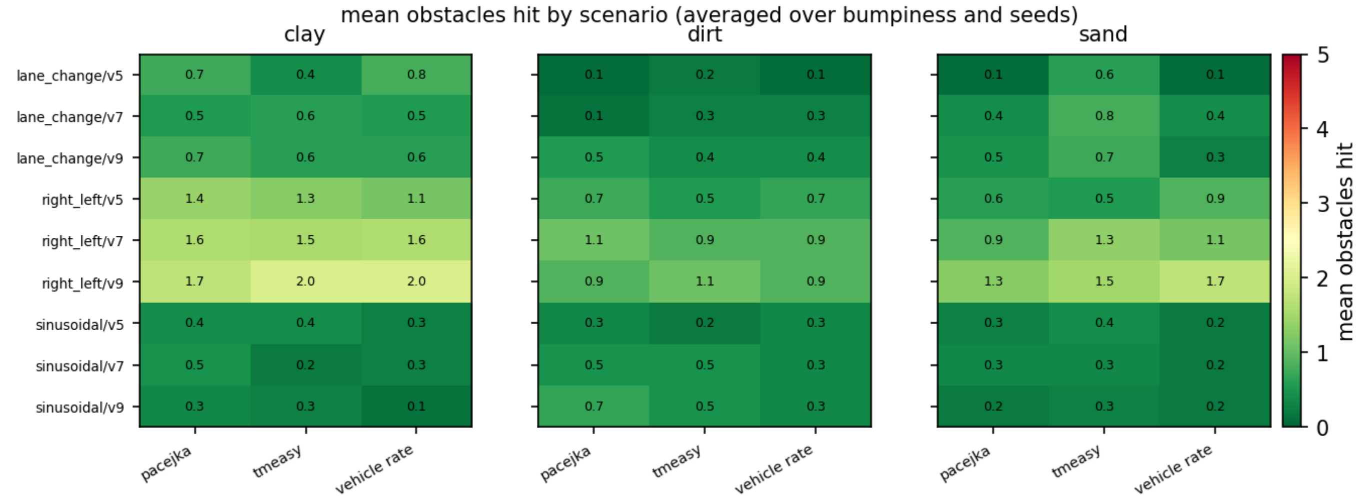

Fig. 19: Autonomous obstacle avoidance, planner tire model swept under a fixed DOB-CBF filter. Mean unique obstacles hit per run stratified by terrain with across-seed whisker error bars; the three tire models are statistically indistinguishable here because the safety filter, not the planner-tire choice, dominates the safety outcome. Sweep matrix is clay/dirt/sand $\times$ 3 paths $\times$ 3 speeds $\times$ 3 bumpiness $\times$ 5 seeds. Aggregate KPIs are in Table XV.

is this paper's most direct filter-attribution result. Because Chrono replay is deterministic, a fixed command trace is replayed through the *identical* scenario with the filter off and on, so every outcome difference is attributable to the filter alone in a paired, reproducible collision-prevention measure with no path-tracking proxy. To avoid resting the result on one input, the trace is drawn from a small *population of scripted worst-case commands*: a "drive-into-the-hazard" command at three throttle levels (0.4, 0.6, 0.8), a stand-in for an inattentive or latency-overwhelmed operator. Table XVI aggregates these intents across five convoy scenarios (lead-brake, 3- and 5-vehicle convoys, rear-approach, stalled) $\times$ three command delays (0, 0.15, $0.30\,\mathrm{s}$) for 45 unfiltered baseline cells and 44 completed DOB-CBF replays. The unfiltered intent collides in 40 of 45 cells; DOB-CBF *prevents* 31 *of the* 39 *matched baseline collisions*, cutting the collision rate from 89% to 20% and turning a mean clearance of $-1.69\,\mathrm{m}$ into $+1.94\,\mathrm{m}$. It does so at small command deviation ($|\Delta\delta| = 0.13$, $|\Delta\tau| = 0.28$), mixing steering with braking — the obstacle-type-aware QP steers around static rocks but is willing to brake behind a vehicle.

The same machinery replays a *recorded* G29 trace unchanged, so logged human intent is a drop-in input to this protocol. We replayed ten such drives — the five convoy scenarios under both a good and a congested learned 5G profile (Sec. VIII), the operator driving on the *delayed* camera feed and through the delayed command path (glass-to-glass) — filter-off vs DOB-CBF (Fig. 20). Under this genuinely hard condition the operator collided in five of the ten drives; DOB-CBF cut that to two, a net *three of five averted*, and lifted the median minimum clearance from 0.15 to $4.42\,\mathrm{m}$ at a modest command deviation ($|\Delta\delta| \approx 0.04$, $|\Delta\tau| \approx 0.15$, below the scripted-intent case and near-inert on the good link). It prevented *all four* forward-collision cases the operator hit — both convoys, the congested lead-brake, and the congested platoon ($-0.69 \rightarrow +10.3\,\mathrm{m}$). Its two residual collisions mark the filter's limits, which we report rather than hide: a rear-approach, which a forward-collision filter cannot prevent by construction (the operator hit it too), and one congested stalled-vehicle case where the filter, rather than committing to the operator's wide lateral bypass, braked the truck to a near-stop ($|\Delta\tau| = 0.37$). The braking is itself a safe response; the resulting $0.5\,\mathrm{m}$ contact is largely an artifact of open-loop counterfactual replay — with the filter running $5\times$ slower than the operator's recorded drive, the operator's post-pass *steering*, replayed on the still-approaching slowed vehicle, drifts it laterally into the stalled vehicle's corner (a closed-loop operator would see the stall and adapt). It nonetheless marks two real limits we report rather than hide: the filter is over-conservative on a stationary blocker under heavy delay (it stops rather than swerving), and open-loop command replay loses fidelity when the filter substantially alters the speed profile. On genuine human intent the filter thus restores large clearance margins on the cases it is designed for, without fighting a capable driver. (Single operator; a multi-operator study is left to future work.)

G. Real-time cost

DOB-CBF is a single small quadratic program solved once per control tick. Against the $100\,\mathrm{ms}$ ($10\,\mathrm{Hz}$) shield budget its solve scales benignly with the number of in-range obstacles N (a rock and a traffic vehicle are identical (x, y, r) entries, so N counts both): a fraction of a millisecond at $N = 0$ rising to only a few milliseconds at $N \sim 100$. It therefore never approaches the budget and imposes no practical obstacle-count limit — one reason it is well suited to the dense boulder-and-convoy scenes here.

TABLE XVI: Counterfactual convoy replay: a population of scripted worst-case commands (constant throttle $\in \{0.4, 0.6, 0.8\}$ into the hazard) replayed filter-off vs DOB-CBF across 5 traffic scenarios $\times$ 3 command delays (45 unfiltered cells, 44 completed DOB-CBF replays). The paired deterministic replay isolates collision prevention by the filter for these scripted intents. $|\Delta\delta|$, $|\Delta\tau|$ are the mean steering/throttle corrections.

Filter	Coll. rate	Prevented	Clear. (m)	$\lvert\Delta\delta\rvert$	$\lvert\Delta\tau\rvert$
None	0.89 (40/45)	—	-1.69 ± 1.15	—	—
DOB-CBF	**0.20** (9/44)	**31/39**	$\mathbf{+1.94\pm2.10}$	0.13	0.28

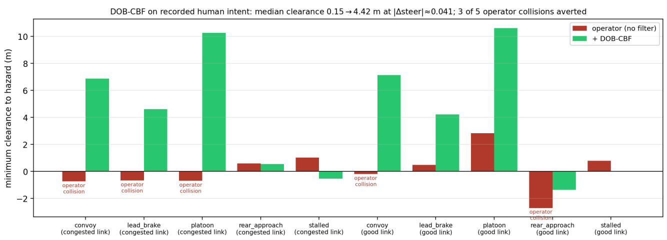

Fig. 20: Counterfactual replay on ten *recorded human* G29 drives (five convoy scenarios $\times$ good/congested 5G latency, driven glass-to-glass on the delayed camera + command paths), operator (no filter) vs DOB-CBF. Under the hard delayed-video condition the operator collides in five of ten (clearance < 0); DOB-CBF averts the four *forward*-collision cases and lifts the median minimum clearance from 0.15 to 4.42 m at $|\Delta\delta| \approx 0.04$. Two residual collisions bound its scope: a rear-approach (a forward filter cannot prevent it) and one congested stalled case where the filter braked to a near-stop instead of swerving wide; that 0.5 m contact is largely an open-loop-replay artifact (operator steering replayed on the 5×-slower filtered vehicle), not a braking failure.

VIII. 5G-Profile Latency Robustness

A sequence model (N-HiTS) is trained on the public 5G YouTube uplink dataset of Choi et al. [2] and used to generate a synthetic bitrate trace. The trace is mapped through a configurable queue-load model to produce a per-channel latency profile that is applied as an additive scheduled delay on the camera and command channels.

The closed-loop obstacle scenario is then run under the synthetic profile with and without the safety filter; Table XVII and Fig. 22 report the result. Under the learned 5G profile DOB-CBF cuts collisions by 86 % (2.85→0.39 per run) and turns a negative mean clearance (−1.48 m, inside the obstacle footprint) into +0.35 m, intervening on about half of the steps. The minimum-deviation, intent-preserving behaviour that suits the human-in-the-loop case (Sec. VII) thus holds up even under the profile's outage spikes.

A. *Does delay*-awareness *matter, or just having a filter?*

The result above shows a filter helps under latency, but not that the filter's *delay compensation* — inflating its obstacle buffer by a delay-dependent stopping distance and forward-predicting over the round trip before acting — is what helps, rather than merely screening the command at all. We isolate it with a deterministic ablation: the command path is delayed by Δ_{cmd} in *every* run, but the filter is either *told* the delay ($\texttt{teleop\_delay} = \Delta_{\mathrm{cmd}}$, "aware") or not ($\texttt{teleop\_delay} = 0$, "blind"); everything else is identical, and the same scripted worst-case commands (Sec. VII-F) are replayed, so blind-vs-aware is paired. We sweep it as a dose-response over six command delays (0 to 0.5 s) $\times$ 5 convoy scenarios $\times$ three intents (90 cells per variant). Both filters reduce collisions substantially relative to no filter (0.87), but the delay-aware filter is strictly better overall: a 0.17 collision rate vs the delay-blind 0.26, at twice the clearance margin (+2.34 vs +1.16 m), preventing 8 of the blind filter's collisions on matched cells. Crucially the gap *grows with delay* (Fig. 23): at $\Delta_{\mathrm{cmd}} = 0$ the two are identical (0.33 each — the sanity check, since with no delay there is nothing to be aware of), by 0.30 s the delay-blind filter still collides at 0.20 while the aware filter is down to 0.07, and by 0.50 s awareness removes the residual entirely (0.20 → 0.00). Intrusiveness is comparable either way ($|\Delta\delta| \approx$ 0.10–0.14), so the gain is smarter timing, not more aggression. This supports the *latency-aware* claim specifically, rather than only showing generic latency robustness.

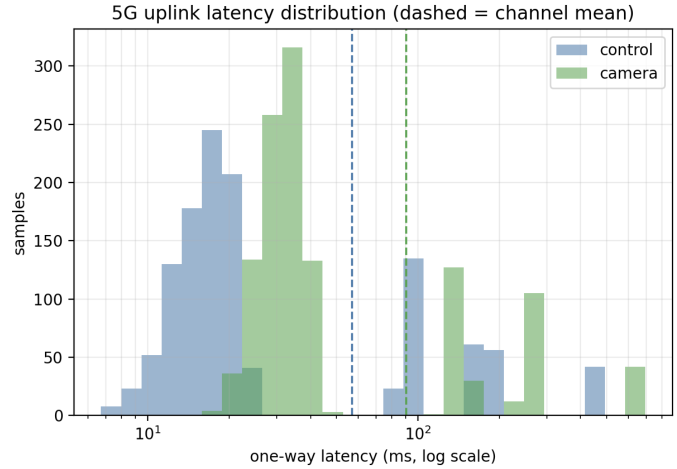

Fig. 21: Learned N-HiTS 5G uplink profile applied to the camera and control channels (log-x histogram; dashed lines are channel means). The profile is bimodal: a low-latency bulk (control around 10–25 ms, camera around 30–50 ms), a higher-latency cluster near 100–200 ms, and discrete outage spikes at 450 ms (control) and 660 ms (camera). Mean one-way latency is approximately 57 ms (control) and 91 ms (camera).

TABLE XVII: Closed-loop collisions under the learned N-HiTS-5G latency profile (DOB-CBF vs no filter).

Filter	Coll./run	Clearance (m)	Interv. (%)
None	2.85 ± 1.14	-1.48 ± 0.53	—
DOB-CBF	$\mathbf{0.39\pm0.58}$	$\mathbf{+0.35\pm1.02}$	51

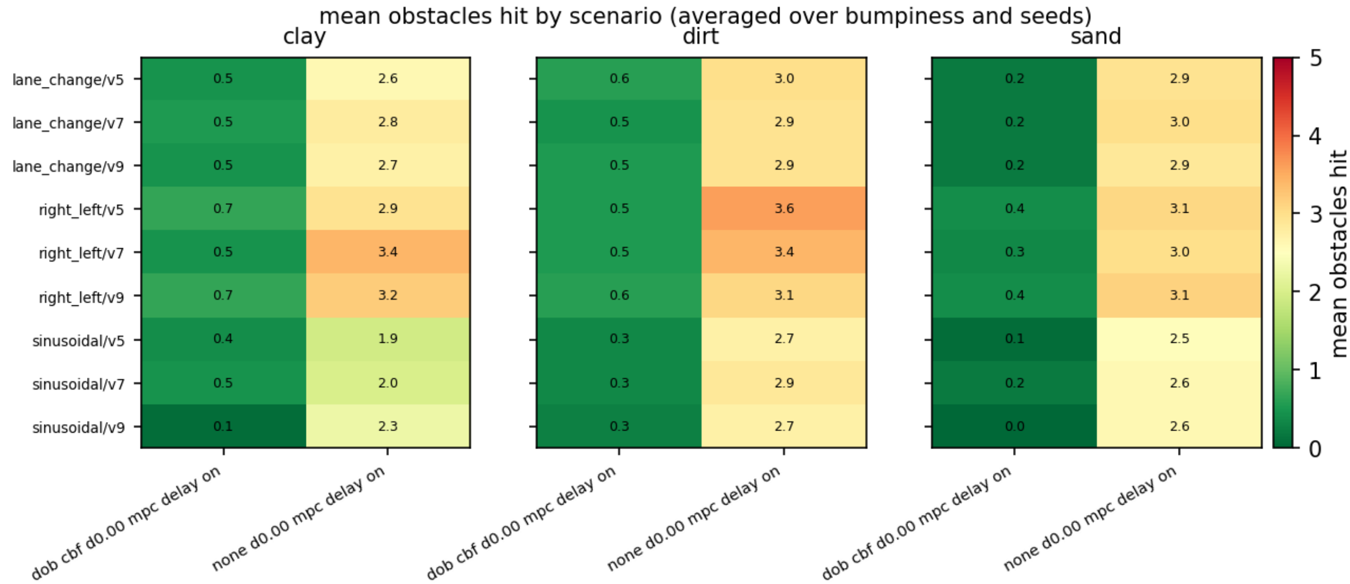

Fig. 22: 5G-profile latency compensation sweep: per-scenario collisions. Aggregate KPIs are in Table XVII.

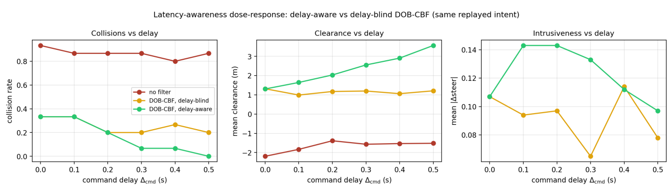

Fig. 23: Latency-awareness dose-response. The same scripted worst-case commands are replayed with the command path delayed by Δ_{cmd}, the DOB-CBF filter either *told* that delay (aware) or not (blind), swept over six delays. At zero delay the two coincide (nothing to be aware of); the gap grows with delay — by $0.5\,\text{s}$ awareness removes the residual collisions the delay-blind filter still suffers, at comparable intrusiveness. 90 deterministic cells per variant.

IX. Terrain- and Latency-Aware Collision Warning

The safety filter in §VII–§VIII modifies the operator's command stream when a collision is imminent. In some deployment modes (e.g. expert-loop teleoperation, or a safety filter that has been muted for debugging) the operator wants only a *notification* rather than a control override. We add an independent forward collision warning component that runs in parallel with the rest of the stack and emits a discrete severity signal $\{\text{GREEN}, \text{YELLOW}, \text{ORANGE}, \text{RED}\}$ on every controller tick.

The warning module is exposed through a small factory in line with the rest of the swappable safety layer, so the warning strategy can be selected at construction time. The default strategy is a time-to-collision predictor with two adjustments:

a) Terrain awareness.: The required stopping distance $d_{\text{stop}} = v\tau_r + v^2/(2a_b)$ uses a brake-deceleration $a_b(\hat{n})$ that is computed *analytically at init time* by querying the same rig tire surrogate the DOB-CBF and MPC use for traction-budget queries. For every $\hat{n} \in [0.40, 1.30]$ on a 0.05 grid, the surrogate is loaded with the manifold-interpolated Bekker vector at that $\hat{n}$, swept over braking slip ratios $\kappa \in [-0.40, 0]$ at static per-wheel F_z, and the peak $|F_x|$ at front and rear axles is summed to give the total available braking force; $a_b(\hat{n}) = \sum F_{x,\text{peak}}/m_{\text{veh}}$. The resulting table runs from $a_b(0.4) \approx 2.3\,\text{m/s}^2$ (soft, clay-like) to $a_b(1.1) \approx 4.7\,\text{m/s}^2$ (firm, sand-like) — both endpoints noticeably more conservative than the hand-tuned $2.5/6.0\,\text{m/s}^2$ heuristic the implementation previously used. The live $\hat{n}$ from the online terrain estimator (§IV) indexes into this table at runtime, so the warning is more conservative on soils that the vehicle cannot brake hard on.

b) Brake-decel validation.: The analytical $a_b(\hat{n})$ table is validated against actual Chrono SCM behaviour with 27 full-brake stops: three terrains $\times$ three initial speeds $\{3, 5, 7\}\,\text{m/s}$ $\times$ three seeds, each scenario cruising to target speed and then applying $\text{brake} = 1$, $\text{throttle} = 0$ until the vehicle is below $0.2\,\text{m/s}$. Stopping distance and mean decel are extracted from the recorded chassis trajectory and compared to the prediction at the brake-onset speed (Fig. 24). On mean decel, the rig-NN analytical table is within $0.1\,\text{m/s}^2$ on clay, over-predicts dirt by $0.7\,\text{m/s}^2$ (the classical contact-patch saturation gap of §III-D), and over-predicts sand by $1.1\,\text{m/s}^2$ — small enough that the resulting stopping-distance prediction lands within $\pm 0.5\,\text{m}$ of the recorded ground truth on 23 of 27 trials. Aggregated mean $|\Delta d_{\text{stop}}|$ is $0.17\,\text{m}$ for the analytical table versus $0.45\,\text{m}$ for the hand-tuned linear-interp fallback ($2.6\times$ improvement); the analytical prediction also has a small positive mean bias ($+0.10\,\text{m}$), meaning the warning is biased toward firing slightly earlier than strictly necessary, which is the safe direction.

c) HIL integration.: The warning module is wired through the sim node so any HIL teleoperation round can

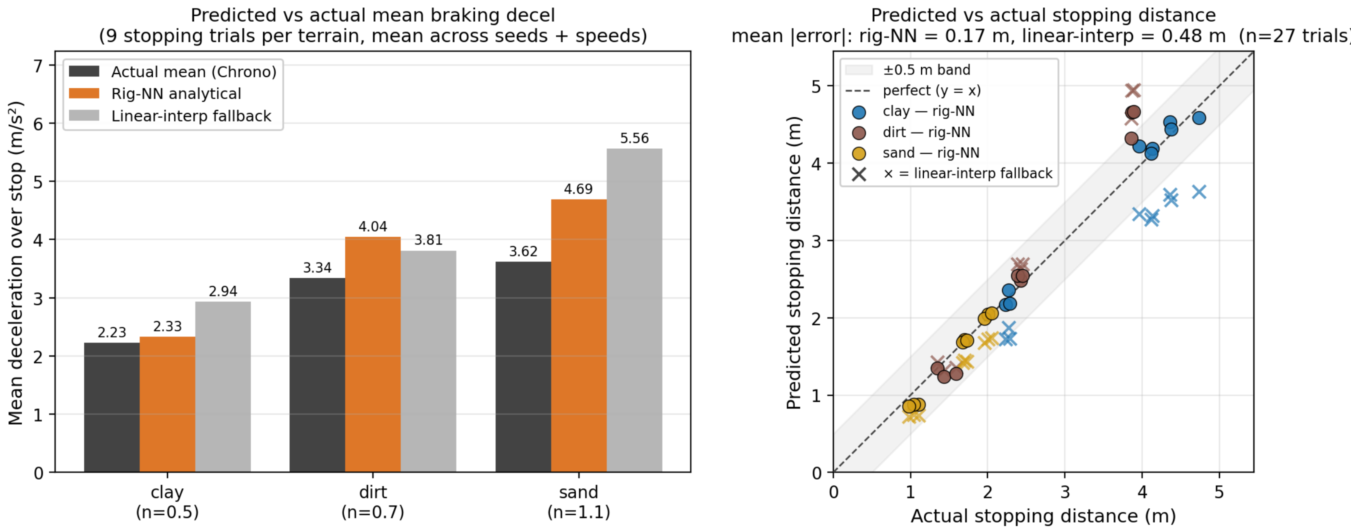

Fig. 24: Validation of the warning system's analytical brake-deceleration table against actual Chrono SCM stopping behaviour (27 trials, 3 terrains $\times$ 3 initial speeds $\times$ 3 seeds). Left: mean deceleration over each stop. The rig-NN analytical estimate (orange) matches the Chrono mean (black) closely on clay and slightly over-predicts on dirt and sand; the original linear-interp fallback (grey) under-predicts on clay and over-predicts on sand. Right: predicted vs actual stopping distance with the $\pm 0.5\,\mathrm{m}$ band shaded. Rig-NN points (circles) cluster tighter to $y = x$ than linear-interp ($\times$, scatter); mean $|\Delta d_{\mathrm{stop}}|$ is $0.17\,\mathrm{m}$ analytical vs $0.45\,\mathrm{m}$ linear-interp.

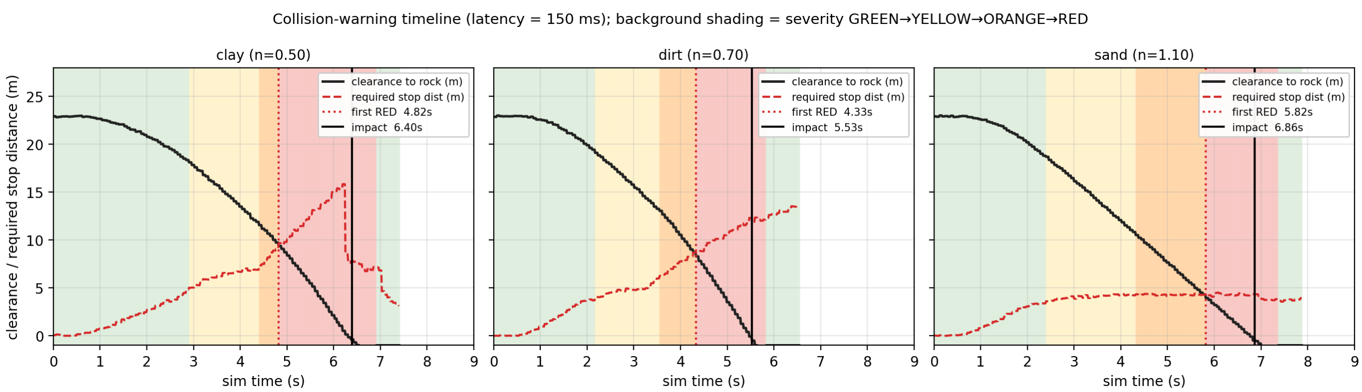

Fig. 25: Severity escalation over time for one representative blind-drive run per terrain at $150\,\mathrm{ms}$ one-way delay. Background shading is the live severity band (GREEN $\rightarrow$ YELLOW $\rightarrow$ ORANGE $\rightarrow$ RED); curves are the chassis clearance to the rock (solid) and the warning system's running stopping-distance estimate (dashed). The first RED fires when the clearance curve drops to the stopping-distance curve.

enable it. Per-tick severity is logged alongside the existing diagnostics, and severity transitions are reported on the sim-node console and, when the Irrlicht visualiser is active, surfaced to the operator as a text banner. When the online terrain estimator is active, the live $\hat{n}$ is forwarded to the warning system on every control message so that the brake-decel estimate adapts to the estimator's current terrain belief; otherwise the warning falls back to the nominal n of the configured terrain preset. The warning never modifies commands, so it composes freely with the safety filter (none or DOB-CBF) used in the same round.

d) Latency awareness.: An EMA tracker of recent one-way network delays and a sliding-window jitter estimator inflate the operator reaction time by $\tau_r + \mathrm{RTT} + k\sigma$, where σ is the standard deviation of one-way delays over the last second of received commands. Increasing the latency baseline or the jitter inflates d_{stop} and fires the warning earlier.

TABLE XVIII: Forward collision warning lead time before impact (RED severity) under terrain $\times$ latency sweep. The vehicle drives blindly at a single $1.2\,\mathrm{m}$ rock with constant throttle. Higher latency and softer soil both increase the lead time, as designed.

	lat=0 ms	lat=150 ms	lat=300 ms
clay ($\hat{n}=0.5$)	1.36 s	1.57 s	1.67 s
dirt ($\hat{n}=0.7$)	1.02 s	1.20 s	1.38 s
sand ($\hat{n}=1.1$)	0.82 s	1.04 s	1.38 s

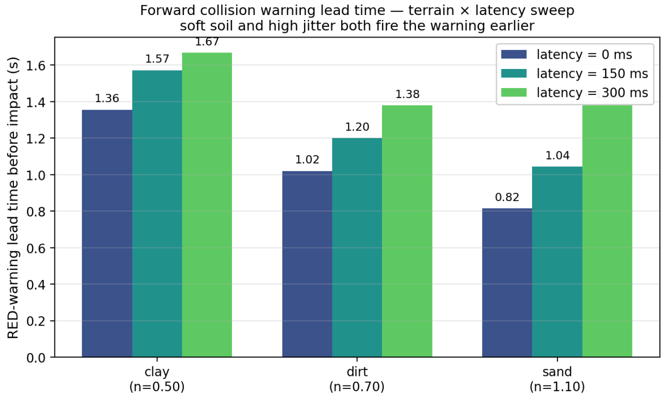

Fig. 26: Forward collision warning lead time before impact in the blind-drive sweep (no MPC, no safety filter, fixed throttle into a single rock). Both the terrain-conditioned brake budget and the latency-conditioned operator-reaction budget enlarge the required stopping distance and therefore advance the first RED warning: soft soil (clay) gets the longest lead, and within every terrain the lead grows monotonically with the configured one-way delay.

The benchmark sweeps three terrains and three one-way latency settings. Table XVIII reports the time between the first RED warning and the rock impact: on soft clay the warning fires approximately 1.36–1.67 s before impact; on firm sand the same vehicle reaches the rock with about 0.82–1.38 s of lead. The lead time also grows with the configured operator latency on every terrain (clay $1.36 \to 1.67$ s as latency goes $0 \to 300$ ms, dirt $1.02 \to 1.38$, sand $0.82 \to 1.38$), confirming that the warning correctly couples both the terrain-conditioned stopping budget and the latency-conditioned operator-reaction budget.

X. Open-Source Benchmarking Suite

A single invocation of the suite's top-level driver executes all of the autonomous sub-sweeps that produce the non-HIL results reported in this paper, then merges the resulting figures and CSVs into the publication set. The paper tier completes over ten thousand Chrono closed-loop trials on a single 24-core workstation; the human-in-the-loop protocol of Sec. VII-A is run separately because it requires an operator at the wheel.

Reproducibility is enforced through a few architectural choices in the suite itself. Sensor noise is enabled in every trial so no oracle quantity is observable at inference time. A per-trial diagnostic CSV is saved alongside each run for post-hoc inspection. Aggregated KPIs deduplicate on the run key (*variant*, *terrain*, *path*, *speed*, *bumpiness*, *seed*), so a re-run cleanly fills only the missing cells without overwriting existing rows. The ZeroMQ publisher retries socket bind with linear backoff to eliminate TIME_WAIT races under rapid run cycling, and the publication step matches result folders by exact timestamp so that closely-named sweep variants are never conflated.

XI. Integrated System Demonstration

The preceding sections evaluate each layer in isolation on shared infrastructure; the contribution, however, is their *integration*. As a capstone we run the entire stack together on a single mission and present it as one timeline (Fig. 27): a sinusoidal reference driven across a clay $\to$ sand soil transition through a rock field, under the learned 5G command/camera latency, with the terrain-aware NMPC, the online Fused-UKF terrain estimator, the g–g speed planner, the throttle DOB, the DOB-CBF safety filter, and the forward collision warning all active simultaneously. The online estimate $\hat{n}$ adapts across the soil boundary (top); the achieved speed tracks the grip-limited g–g reference with the DOB closing the soft-soil gap (middle); and the safety filter holds the ego clear of the hazards throughout (minimum clearance 0.42 m, 162 binding-constraint avoidance steps) with *zero collisions* over the 129 m traverse (bottom). Running the full stack is itself an integration test: it exercises the cross-layer interfaces that per-component evaluation does not — here the estimator's live-terrain update re-conditions the safety filter's grip-limited braking authority, the same socket the controller uses to re-condition its own surrogate. The run is reproduced by `benchmarking/integrated_hero_run.py`. This single run is an integration check rather than a separate statistical safety claim.

XII. Discussion

Two recent threads are directly adjacent to this framework and merit explicit comparison. Onozuka et al. [9] replace a terrain-parameter-dependent NN tire model with a physics-informed structure – a Janosi saturating shear law wrapped around two small networks that supply the force coefficients – and re-fit those coefficients online by stochastic gradient descent on state-derivative residuals. The approach is complementary to ours: they fold soil mechanics into a structured force law where we estimate the soil and re-condition an unstructured surrogate. A direct adaptation of their tire model into the per-axle interface of our acados solver improved offline force prediction (held-out $R^2_{F_y}$ from 0.35 to 0.59) but did not run stably in *our* in-horizon NMPC, where the static formulation triggered QP failures on aggressive transients. We stress this is *not* a fair head-to-head: their model is designed for the dedicated curvilinear-coordinate, wheelspeed-state MPC of [9], which we did not reimplement, so we draw *no* superiority conclusion from it — only that the two approaches are complementary (they structure the force law; we estimate the soil and re-condition an unstructured surrogate around it). Gupta et al. [10] compensate unmodeled dynamics with a reinforcement-learning residual on top of MPC; in a feasibility prototype on the same plant the residual closed less than 8 % of the residual tracking error and only at the largest model mismatch – consistent with our own earlier finding that a learned dynamics residual offered little benefit on this plant – so this paper does not adopt RL compensation.

The empirical evidence supports the central integration claim and the principal component choices. The whole-vehicle

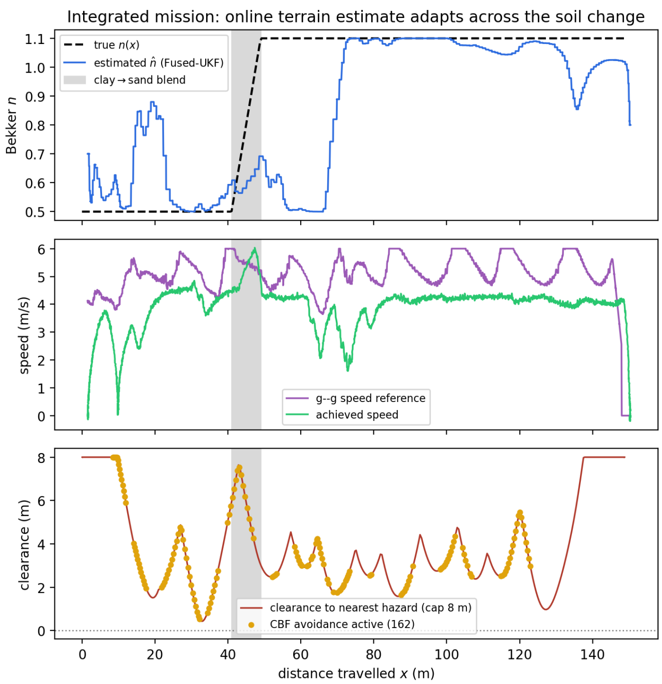

Fig. 27: Integrated mission: the whole stack on one run, under the learned 5G latency, across a clay $\rightarrow$ sand soil change (grey band) through a rock field. **Top:** the online terrain estimate $\hat{n}$ (blue) tracks the true $n(x)$ (dashed) across the soil boundary. **Middle:** achieved speed (green) tracks the terrain-aware g–g reference (purple). **Bottom:** clearance to the nearest hazard (capped at 8 m) dips at each rock, with the DOB-CBF binding-constraint avoidance steps marked; zero collisions over the traverse.

neural tire surrogate averages $28\,\mathrm{ms}$ in the static tire-model sweep and $32\,\mathrm{ms}$ in the live-estimator sweep (both within the $100\,\mathrm{ms}$ MPC step) and reduces median RMS crosstrack error against the *SCM-calibrated* analytical baselines by 37–38 % in the live-estimator regime (Table IV), separating clearly from them in the tail across the full terrain matrix (Table III), where even the calibrated baselines diverge on off-nominal dirt. The asymmetric throttle disturbance observer increases achieved speed by approximately 5 % ($4.07 \rightarrow 4.26\,\mathrm{m/s}$, $0.61 \rightarrow 0.64$ of the g–g profile) at a small ($\sim 7\,\mathrm{mm}$) tracking cost. The two-layer planner-aware stack drives collisions to 0.13/run with the DOB-CBF filter, the DOB-CBF filter benefits substantially from re-using the neural surrogate, and a deterministic counterfactual replay over a population of scripted worst-case commands attributes the averted collisions to the filter (31 of 39, collision rate $89\% \rightarrow 20\%$) in a paired, reproducible safety measure that avoids human-operator variance. The forward collision warning closes the advisory gap the override-style filter leaves open: its analytical stopping-distance prediction lands within $0.17\,\mathrm{m}$ of the actual Chrono stop ($2.6\times$ better than the prior hand-tuned fallback), and lead time grows monotonically with both softer soil and longer one-way delay.

The terrain estimator is intentionally scoped to the sinkage exponent n, which is the parameter that transfers cleanly under closed-loop sinusoidal excitation. The HIL protocol of Section VII-A is implemented with a Logitech G29 and delayed command/camera channels; the autonomous sweeps provide the repeatable quantitative evidence, and any session can additionally enable the forward collision warning alongside the chosen filter.

A. Limitations and scope

This study is conducted entirely in simulation, on a single HMMWV plant on Chrono's Soil Contact Model. We make no sim-to-real transfer claim: the contribution is the modular, latency-aware *architecture* and the relative comparisons it enables on a high-fidelity, widely-used deformable-terrain plant [4], [7]. Those comparisons — learned surrogate versus analytical tire model, safety filter versus no safety filter, estimator-on versus wrong-prior — are internal to one plant and one harness, so they are unaffected by any absolute sim-to-real offset; and the plant is itself an extension point, so porting to hardware or another terramechanics model is a registration rather than a rewrite. The neural surrogate is trained and evaluated on the same SCM plant, so the analytical baselines are calibrated to that plant's mean lateral friction with a single global parameter set (Sec. III-B): the comparison is against a fair, SCM-calibrated analytical model rather than an uninformative rigid-road baseline, and the residual gap reflects what a fixed parameterization structurally cannot capture about terrain-varying deformable forces, not a tuning deficit.

A characterized floor within the surrogate is the rear-axle lateral force. Section III-F shows that, unlike the front axle, rear F_y does not respond to any causal MPC-available feature we tried — dynamic vehicle state, kinematic load transfer, per-wheel sinkage (the multi-pass effect, tested even with ground-truth SCM sinkage as an oracle), or a $0.32\,\mathrm{s}$ slip/state history — all of which leave it within 4% of the deployed model. We therefore treat rear F_y as irreducible at the deployed capacity rather than a tuning target. Separately, the longitudinal channel's soft-soil speed bias is a dynamics-level effect, not a learnable tire feature (F_x RMSE moves at most 3% under the same feature sweep). In the SCM contact model the net longitudinal force is two mechanisms [4]: a sinkage-driven *compaction resistance* (the horizontal projection of the Bekker normal stress on the deformed rut, present even at zero slip) plus a slip-driven Janosi shear traction — and the kinematic prediction model ($\dot{u} = a_x$) carries neither. We compensate the net bias online with the throttle disturbance observer (Sec. VI); a feedforward term for the slip-independent compaction resistance in the prediction model is the natural future replacement, while the slip-dependent part settles within a contact-patch transit and so is well approximated by the static surrogate at the control rate.

The online estimator recovers only the sinkage exponent n and reconstructs the remaining five Bekker–Mohr parameters from $\hat{n}$ along the clay–dirt–sand preset manifold — the parameter-known setting also used by the reference terramechanics estimators (Sec. IV-B). This assumes deployment soils lie near that manifold; soils with large independent excursions in cohesion, friction, or stiffness fall outside the present scope, and n-recovery degrades off the manifold.

The safety filter is fundamentally *forward*-collision-avoidance: its barriers guard the ego's path ahead, and a threat from behind is the harder case, since escaping it requires accelerating or changing lanes rather than the braking/steering the minimum-deviation formulation favors. We added a rear-approach guard to DOB-CBF (it refuses deceleration and accelerates to escape a closing in-lane rear vehicle); on a stationary-ego rear-approach test it accelerated away and cut the impact penetration from 3.3 to $0.6\,\mathrm{m}$, but it cannot guarantee avoidance — a fast tailgater is unescapable from low speed — and this behaviour has so far been validated only on a single rear-approach scenario in simulation. A lane-change escape and a predictive rear cost are the natural extensions. Two caveats bound the safety evaluation specifically. First, DOB-CBF is measured against a *no-filter* baseline rather than competing safety filters: whether the disturbance observer and neural traction budget improve on a textbook CBF-QP [12] or a reachability-based filter [13] is left to future work; the NN-vs-no-NN ablation (Sec. VII-D) isolates one component but not the CBF-QP itself. Second, the human-intent evaluation uses a single operator, and the counterfactual replay loses fidelity when the filter substantially alters the speed profile — in one congested stalled-vehicle case the filter braked to a near-stop (a safe action) but the operator's faster-drive steering, replayed on the slowed vehicle, drifted it into a $0.5\,\mathrm{m}$ contact that is partly a replay artifact rather than a filter failure (Sec. VII-F). A competitive safety-filter benchmark and a multi-operator study are the natural next steps. Estimating the full soil vector online and hardware transfer are the principal directions for future work; the HIL pipeline additionally enables a human-factors teleoperation study, which is a separate undertaking beyond this systems paper.

XIII. Conclusion

We presented a modular, latency-aware framework that integrates terrain-adaptive control, online soil estimation, two-layer obstacle avoidance, and a terrain- and latency-aware collision warning around a single deformable-terrain plant. The novelty is the *integration*: where prior work treats terrain adaptation and delay-compensated safety in isolation, here a differentiable SCM tire surrogate, an online n-estimator that reconditions it, swappable safety filters, and an advisory warning share one plant, one messaging substrate, and one benchmarking harness, so the main cross-layer choices are measured on controlled, comparable matrices. Empirically, the surrogate-plus-estimator recovers near-oracle tracking with no prior terrain knowledge — within $\approx 3\,\mathrm{mm}$ median RMS crosstrack error of the matched-prior oracle and well below SCM-calibrated analytical baselines; the deployed proprioceptive-fused UKF tracks n across the deployable range at $4.5\,\%$ median error; the DOB-CBF safety filter cuts collisions by $86\,\%$ under a learned 5G latency profile, with a deterministic counterfactual attributing 31 of 39 matched averted collisions to it; and the collision warning predicts stopping distance to $0.17\,\mathrm{m}$. The modular extension points make each of these a replaceable component rather than a fixed pipeline, which is what positions the framework for the hardware-in-the-loop and full-soil-estimation extensions above.

Reproducibility

The framework, the swappable-component registries, and the full benchmarking suite are released as open source at https://github.com/ANON/scm-shared-control (anonymized for review). The complete autonomous sweep, including the non-HIL figures and CSV-backed tables in this paper, is reproduced by the following single command in the provided conda environment:

```
python benchmarking/run.py -tier paper
```

The counterfactual safety result (Sec. VII-F), the latency-awareness ablation (Sec. VIII-A), and the integrated mission (Sec. XI) each reproduce from a single self-contained script in `benchmarking/`:

```
python benchmarking/convoy_counterfactual_eval.py
python benchmarking/latency_awareness_ablation.py
python benchmarking/integrated_hero_run.py
```

The exact result folders backing each figure are catalogued alongside the published figures, and a table-value provenance sheet maps each paper table to its backing CSV, script, or code location. The HIL rounds are intentionally excluded from the automated commands because they require a human operator; they are run through a separate human-in-the-loop script.

Acknowledgements

This work builds on the Chrono::HIL framework [11] and Project Chrono [7], [5]. The 5G traffic profile is generated using the public N-HiTS-5G implementation released with Choi et al. [2].

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